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Selenium detoxification is required for cancer-cell survival

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The micronutrient selenium is incorporated via the selenocysteine biosynthesis pathway into the rare amino acid selenocysteine, which is required in selenoproteins such as glutathione peroxidases and thioredoxin reductases[1,2]. Here, we show that selenophosphate synthetase 2 (SEPHS2), an enzyme in the selenocysteine biosynthesis pathway, is essential for survival of cancer, but not normal, cells. SEPHS2 is required in cancer cells to detoxify selenide, an intermediate that is formed during selenocysteine biosynthesis. Breast and other cancer cells are selenophilic, owing to a secondary function of the cystine/glutamate antiporter SLC7A11 that promotes selenium uptake and selenocysteine biosynthesis, which, by allowing production of selenoproteins such as GPX4, protects cells against ferroptosis. However, this activity also becomes a liability for cancer cells because selenide is poisonous and must be processed by SEPHS2. Accordingly, we find that SEPHS2 protein levels are elevated in samples from people with breast cancer, and that loss of SEPHS2 impairs growth of orthotopic mammary-tumour xenografts in mice. Collectively, our results identify a vulnerability of cancer cells and define the role of selenium metabolism in cancer.

Altered metabolism is a hallmark of cancer[3], and enzymes that produce metabolites required by cancer cells are attractive targets for therapy. However, targeting these enzymes may have limited efficacy when the same metabolites are available in the tumour environment, or can be produced through alternative routes[4]. As an alternative therapeutic approach, we sought to identify metabolic enzymes that may be required in cancer cells to prevent the buildup of toxic metabolites (Fig. 1a). To first identify endogenously produced toxic metabolites, we cross-examined the metabolites in the human metabolic network with toxicological data from the NIH Toxicology Data Network (TOXNET)[5]. The enzymes that use these toxic metabolites as substrates according to the Kyoto Encyclopedia of Genes and Genomes (KEGG)[6] were designated as putative detoxifying enzymes, and a panel of these were examined for their role in cell viability (Extended Data Fig. 1).

CRISPR–Cas9-based gene disruption[7] of several of these enzymes impaired cell viability in one or more of the tested cell lines (Fig. 1b). Loss of SEPHS2 was particularly detrimental, causing a loss of viability of over 40% in 6 of the 11 cell lines tested, including the breast-cancer line MDAMB231 and the glioma line U251 (Extended Data Fig. 2a). We designated SEPHS2 as a candidate detoxifier because its annotated substrate is selenide (H_2Se), a compound that has been reported to be toxic in accidental industrial- or laboratory-exposure cases[8,9]. After expanding our testing to additional cancer lines and to non-transformed cells, we found that SEPHS2 knockout (KO) was toxic to 12 of 22 cancer cell lines but none of the 7 tested normal lines (Fig. 1c and Supplementary Table 2). SEPHS2 KO in cancer cells induced loss of cell proliferation (Extended Data Fig. 2a), loss of colony-forming capacity (Fig. 1e) and cell death (Extended Data Figs. 2b and 3a–d). Notably, loss of SEPHS2 was non-toxic to all normal (non-transformed immortalized or primary) lines that we tested, such as the mammary epithelial lines MCF10A and MCF12A, as well as primary lung and colon fibroblasts (Fig. 1d), despite efficient KO (Extended Data Fig. 9a). Furthermore, MCF10CA1h cells, which are HRAS-transformed, phenotypically malignant versions of MCF10A cells[10], are sensitive to SEPHS2 KO (Extended Data Figs. 2c and 9a), providing an isogenic model for SEPHS2 dependency in transformation and malignant progression. Importantly, overexpression of a mutant form of SEPHS2 (SEPHS2-U60C) that is resistant to one of the guide RNAs (guide 1, g1) rescued cells from the toxic effects of SEPHS2 KO from the corresponding gRNA, demonstrating that the observed toxicity was not due to off-target effects of CRISPR (Fig. 1e and Extended Data Fig. 2d,e). Furthermore, we verified that SEPHS1 does not carry out an analogous function to that of SEPHS2 (Extended Data Fig. 2i,j), as has been previously indicated[11]. Thus, SEPHS2 is an enzyme that is essential for survival in many cancer cells, but is surprisingly dispensable in normal cells.

To validate these findings in vivo, we disrupted SEPHS2 in an orthotopic xenograft model for breast carcinoma. Loss of SEPHS2 resulted in a substantial improvement in tumour-free survival of the host (Fig. 1f), which appeared to be due to a combination of reduced tumour formation (Fig. 1g) and average tumour weight and volume (Extended Data Fig. 2f,g). The few SEPHS2-KO tumours that did form showed varying expression of SEPHS2 protein levels, suggesting both an expansion of cells that escaped KO and adaptation of KO cells to allow survival (Extended Data Fig. 2h). Meta-analyses of *SEPHS2* gene expression in cancer and normal tissues revealed that SEPHS2 was overexpressed at the messenger-RNA level across many types of tumours, including breast cancer, relative to its expression in their normal tissue counterparts (Extended Data Fig. 4a–c). SEPHS2 protein was also substantially upregulated in human breast-cancer tissues compared with its that in normal breast tissue (Fig. 1h,i). Notably, SEPHS2 expression was associated with poor survival in people with breast cancer (Fig. 1j). These results demonstrate that SEPHS2 is an enzyme that is both overexpressed in and required for the survival of breast-cancer cells.

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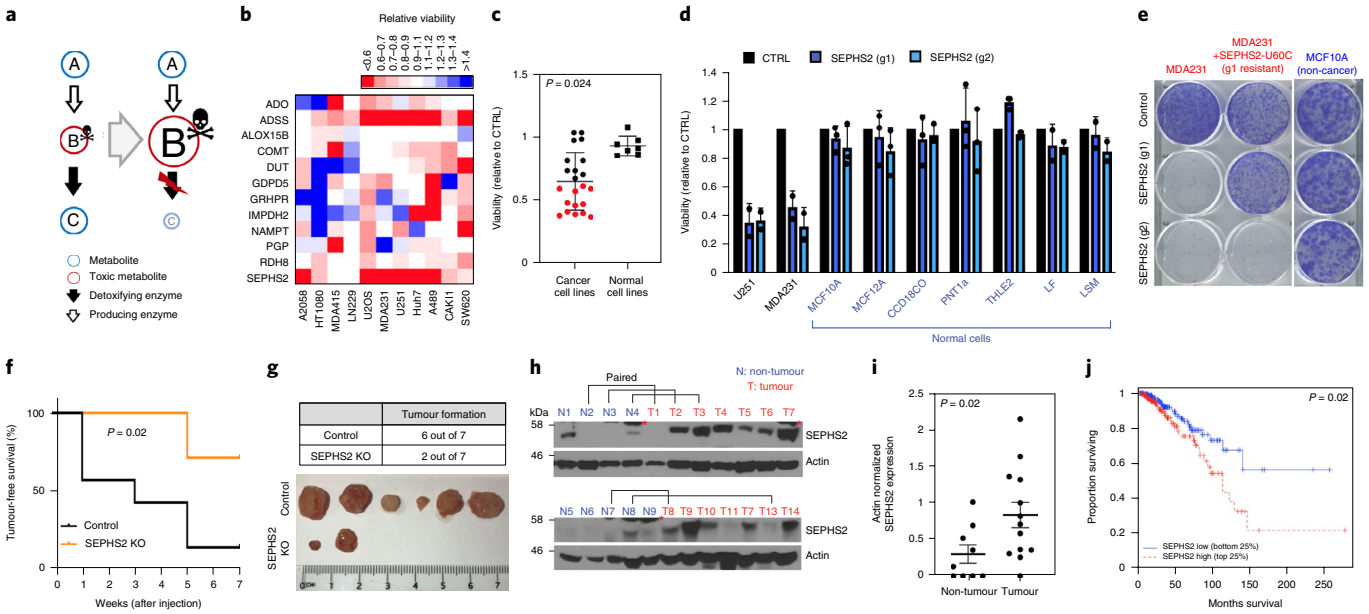


Fig. 1 | Identification of SEPHS2 as an enzyme that is selectively essential to cancer cells and is a potential therapeutic target. a, Toxic-metabolite poisoning strategy. If B is a toxic metabolite, and the downstream (detoxifying) enzyme is disrupted, toxic accumulation of B, rather than the loss of metabolite C, may cause toxicity. **b**, Heat map representing the impact of KO of each putative detoxifying enzyme (rows) on viability (measured as described in the Methods) of cancer cell lines (columns). Values are relative to expression of a non-targeting gRNA in cells from the same line (which was set at 1; see 'Determining relative cell viability following CRISPR–Cas9 KO' for more information); red indicates a decrease and blue indicates an increase in viability. Detoxifying enzyme information is in Extended Data Fig. 1. MDAMB231 is abbreviated to MDA231, and MDAMB415 to MDA415. **c**, Dot plot of effect of SEPHS2 KO on the viability of 22 cancer cell lines and 7 non-cancer cell (non-transformed and primary) lines of varying tissue origin, with each dot representing one cell line. Values are relative to expression of a non-targeting control gRNA (CTRL) in cells from the same line. Detailed information on cell lines and viability data for **b** and **c** are shown in Supplementary Table 2. SEPHS2_g1 and SEPHS2_g2 are two different gRNAs targeting *SEPHS2*. **d**, The viabilities of non-cancer lines following KO, with gRNAs against *SEPHS2* (blue bars), for 9 d. Values are relative to expression of a non-targeting gRNA in cells from the same line (black bars). SEPHS2 KO efficiency for each line is shown in Extended Data Fig. 9. **e**, Crystal violet staining showing the colony-forming ability of MDAMB231 cells, MDAMB231 cells overexpressing g1-resistant SEPHS2-U60C and MCF10A cells, following SEPHS2 KO via g1 and g2. **f**, Kaplan–Meier plot of tumour-free survival for mice orthotopically injected with SEPHS2 KO or control KO MDAMB231 cells. $n = 7$ mice for each condition. **g**, Tumour-formation data and ex vivo images of the tumours at the endpoint. **h**, SEPHS2 immunoblots from cancer tissue and normal breast tissue from people with breast cancer. Paired samples from the same person are indicated. Red asterisks indicate non-specific bands present in mammary tissue. **i**, Quantification of SEPHS2 band intensities from **h**, normalized to that of actin; each dot represents one sample from **h**. **j**, Overall survival estimates for people with breast carcinoma in TCGA based on SEPHS2 expression (SEPHS2 low, $n = 180$; SEPHS2 high, $n = 191$). For **d**, each point is an average from biological triplicates from an independent set. For **c**, **d** and **i**, error bars are s.e.m. *P* values were calculated using two-tailed Student's *t*-test.

Collectively, these findings indicate that SEPHS2 is a highly attractive target for cancer therapy, which may be druggable because SEPHS2 is a metabolic enzyme with hydrophobic substrate pockets[12].

SEPHS2 is required for production of selenoproteins, a group of 25 proteins containing selenocysteine residues, which include antioxidant enzymes such as glutathione peroxidase 4 (GPX4) and thioredoxin reductase 1 (TXNRD1)[1,2,11]. Thus, it was surprising that SEPHS2 was not essential for the survival of normal cells. SEPHS2 phosphorylates selenide to form selenophosphate, which serves as the selenium donor for the biosynthesis of selenocysteinyl transfer RNA by phosphoryl-tRNA kinase (PSTK) and *O*-phosphoserine tRNA-selenocysteine tRNA synthase (SEPSECS) (Fig. 2a). Disruption of any of these enzymes resulted in a dramatic decrease in selenoprotein levels (Fig. 2b), including that of SEPHS2 itself as it is also a selenoprotein. Strikingly, the non-transformed MCF10A cells subjected to SEPHS2 KO maintained their growth rate for 28 d, despite having lost SEPHS2 and the capacity to express selenoproteins (Fig. 2c,d). However, SEPHS2 KO in MDAMB231 cancer cells produced a severe reduction in growth rate, which ultimately recovered around day 20 (Fig. 2c). SEPHS2 and several selenoproteins were initially depleted in the population, but then recovered between days 16 and 20 (Fig. 2d). This indicated that SEPHS2 KO cells were strongly selected against and died out, with the population being replaced by wild-type cells which escaped CRISPR KO. Collectively, these results indicate that normal cells require neither SEPHS2 nor selenoprotein production for their survival and proliferation under basal conditions.

As selenoprotein production by SEPHS2 per se was not essential for cell survival, we considered the hypothesis that cancer cells may have a selenium-detoxification demand that normal cells do not have. Interestingly, in the 1960s, it was observed that radioactive selenite (SeO_3^{2-}), a common dietary form of selenium, preferentially accumulated in tumours, and this was briefly explored as a tumour-labelling strategy[13,14]. Therefore, we examined whether we could recapitulate this selenium uptake in cancer cell lines, as this may determine the detoxification requirement. As excessive selenite is toxic to cells[15], we carried out a focused CRISPR screen to identify genes required for selenium import, assuming that KO of genes necessary for either selenite import or selenide production would prevent selenite toxicity. The 21 candidate genes were selected on the basis of a literature search and on homology from selenium transporters in other species (Extended Data Fig. 5a). KO of SLC7A11 and SLC3A2, the two components of the xCT cystine–glutamate antiporter[16,17], ameliorated selenite sensitivity (Fig. 3a and Extended Data Fig. 5b,c), and SLC7A11 KO prevented selenium accumulation in cells treated with selenite (Fig. 3b).

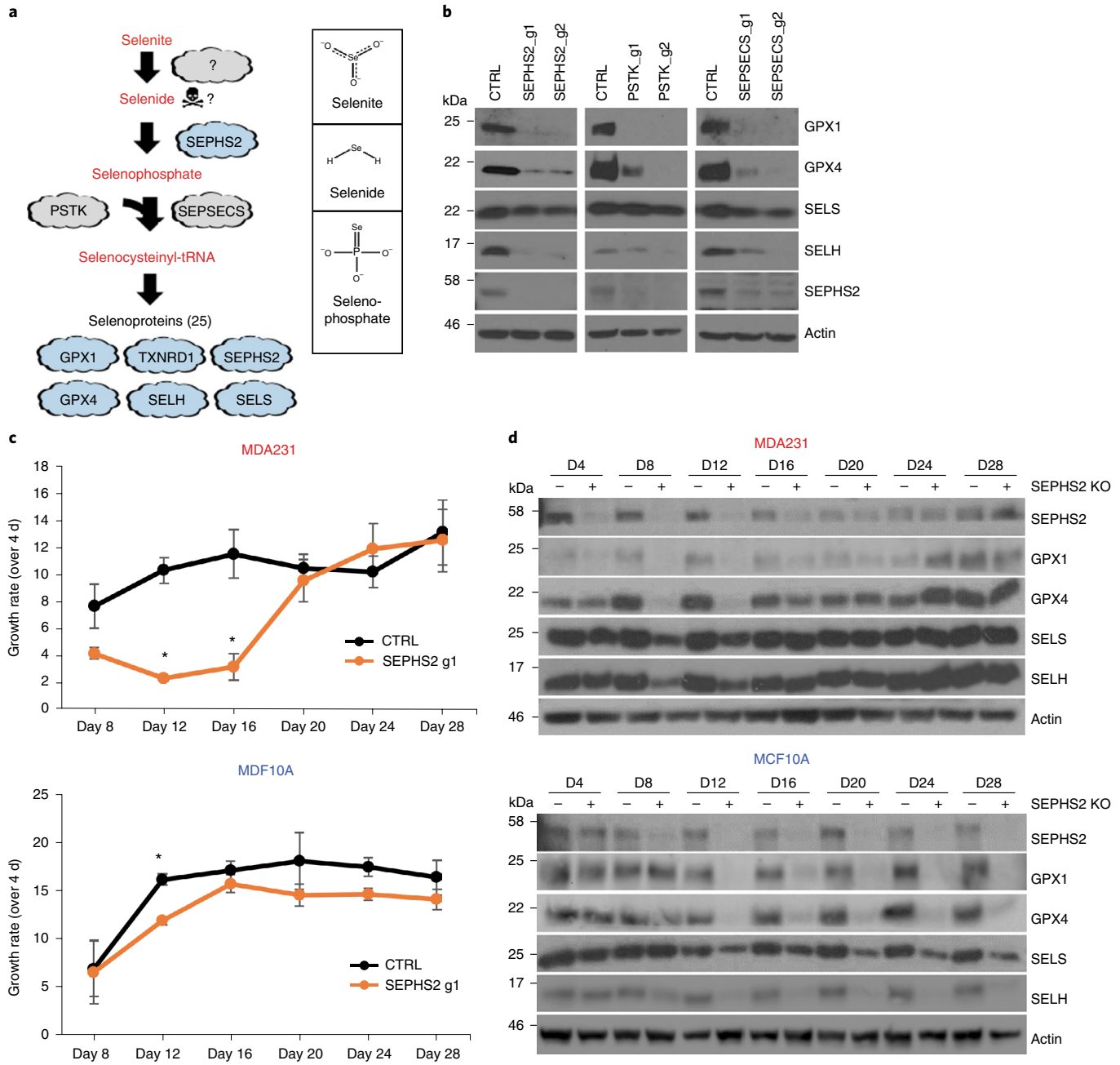


Fig. 2 | SEPHS2 is dispensable to non-transformed cells despite being required to produce selenoproteins. a, Diagram of selenocysteine biosynthesis pathway and chemical structure of selenite, selenide and selenophosphate. **b**, Immunoblots of selenoprotein expression in MDAMB231 cells subjected to CRISPR–Cas9-induced KO with gRNAs against genes encoding enzymes in the selenocysteine biosynthesis, SEPHS2, PSTK and SEPSECS, or control gRNAs for 11 d. **c**, Long-term monitoring of growth rates of MDA-MB-231 cells and MCF10A cells subjected to CRISPR–Cas9-induced KO with gRNAs against SEPHS2 or control, over the course of 28 d. Each point represents the fold change of growth over the previous 4 d. **d**, Immunoblots of selenoproteins expressed by MDAMB231 and MCF10A cells, taken at each time point from Fig. 2c. For **c**, $n = 3$ biological replicates; error bars are s.e.m. *$P < 0.05$ (two-tailed Student's t-test).

In examining a panel of breast-cancer and non-transformed lines, we found that cells that expressed elevated levels of SLC7A11 took up more selenium (that is, they were 'selenophilic') than did normal and cancer cells with little or no SLC7A11 (Fig. 3c,d). Thus, SLC7A11 is required for selenium import and for the toxic effects of selenite in cancer cells.

It has been shown that import of cystine by xCT results in its intracellular reduction to cysteine and subsequent export; this increases extracellular reduction potential in the form of reduced thiol groups from cysteine, which may be involved in selenium uptake[18]. Indeed, selenophilic cells had increased extracellular thiols (Fig. 3e), and this was dependent on xCT (Fig. 3f and Extended Data Fig. 5d,e). xCT allowed selenium uptake via extracellular reduction of selenite, rather than by acting as its physical transporter, as providing extracellular thiols by adding glutathione to medium sensitized cells to selenite toxicity even in the absence of functional xCT (Fig. 3g). We hypothesized that the reduction of selenite to selenide through this mechanism would be sufficient for selenium import, as selenide is highly volatile. The reduction of selenite is key to its toxicity, as we could completely prevent or induce toxicity by modulating

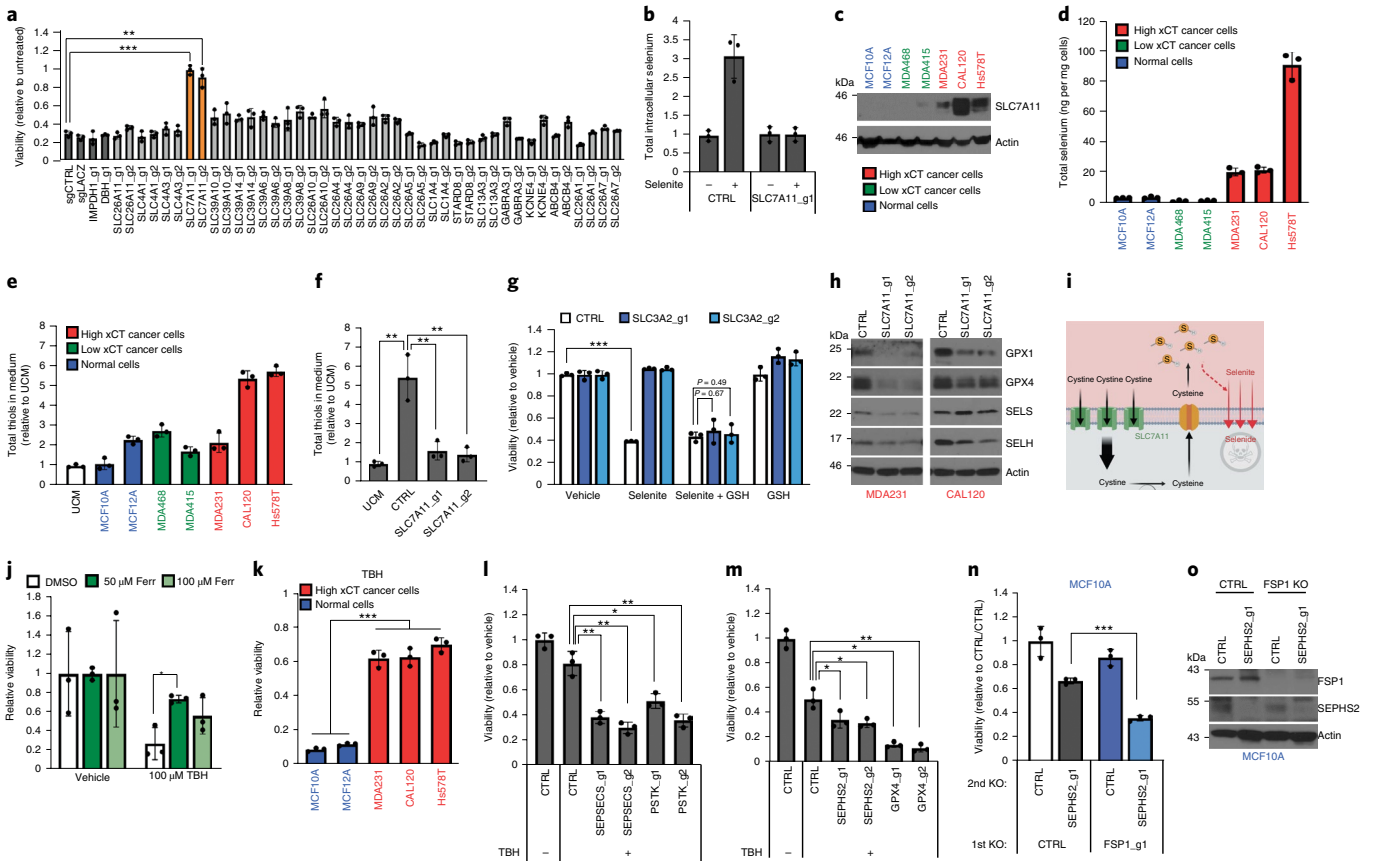


Fig. 3 | SLC7A11-mediated reduction of selenite is the initial step of the selenocysteine biosynthesis pathway in cancer cells, and protects against ferroptosis. a, Viability of U251 cells subjected to KO with gRNAs against various control genes (dark grey bars) or against candidate selenium transporters, and then 48 h of treatment with 12 μM sodium selenite. Values are relative to viability of KO cells from the same line treated with vehicle. **b**, Selenium uptake in U251 cells either with or without SLC7A11 depletion. Total intracellular selenium was quantified following 12 μM selenite supplementation of for 2 h in U251 cells. Values are relative to that in control cells from the same line treated with vehicle. **c**, Immunoblots of SLC7A11 in a panel of breast cancer and normal breast lines. Non-cancer lines are in blue, cancer lines with low SLC7A11 expression are in green and cancer lines with high SLC7A11 expression are in red here and in subsequent histograms. **d**, Selenium uptake in the same panel of cell lines. Total intracellular selenium levels were measured following treatment with 12 μM selenite for 2 h. **e**, Total thiol quantification of media from the panel, conditioned for 24 h. Each value was normalized to that of unconditioned medium (UCM) for every thiol-quantification assay. **f**, Total thiol quantification of media from control KO and SLC7A11-KO MDAMB231 cells that were conditioned for 24 h. **g**, Viability of control and SLC3A2-KO MDAMB231 cells treated with vehicle, 12 μM sodium selenite and/or 75 μM reduced L-glutathione (GSH) for 72 h. Values are relative to viability of cells from the same line treated with vehicle. **h**, Immunoblots of selenoproteins in triple-negative breast-cancer cells subjected to KO with gRNA against *SLC7A11* for 11 d. **i**, Representation of the model revealed by the results in Fig. 3. **j**, Viability of MDA231 cells pretreated with ferrostatin (Ferr) for 3 h and then with vehicle or 100 μM TBH for 48 h. Values are relative to viability of cells from the same line treated with vehicle. **k**, Viability of selenophilic cancer cell lines (red) and non-selenophilic normal lines (blue) treated with 100 μM TBH for 24 h. Values are relative to viability of cells from the same lines treated with vehicle. **l**, Viability of MDA231 cells subjected to KO with gRNAs against *SEPSECS* or *PSTK* for 5 d, then treated with vehicle or 50 μM TBH for 6 h. Values are relative to viability of cells from the same line treated with vehicle. **m**, Viability of MDA231 cells subjected to KO with gRNAs against *SEPHS2* or *GPX4* for 5 d and then treated with vehicle or 250 μM TBH for 24 h. Values are relative to viability of cells from the same line treated with vehicle. **n**, Viability of MCF10A cells subjected to sequential gene KO as shown. Values are relative to viability of cells from the same line subjected to CTRL KO and then to CTRL KO. **o**, Immunoblots of FSP1 and SEPHS2 from **n**. FSP1 expression did not substantially differ between SEPHS2-sensitive lines and SEPHS2-insensitive lines (Extended Data Fig. 6e). For **a**, **d**–**g** and **j**–**n**, $n = 3$ biological replicates; for **b**, $n = 3$ technical replicates; error bars are s.d. For **d** and **e**, although trends can be seen, statistical comparisons were not carried out owing to variability between cell lines. For all panels, the measure of centre is the mean. For all panels, $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ (Student's two-tailed *t*-test).

extracellular thiol levels (Fig. 3a,g). Disruption of SLC7A11 resulted in reduced selenoprotein expression (Fig. 3h), demonstrating its role in promoting selenoprotein production. Collectively, these results show that SLC7A11 is involved in the following model (Fig. 3i): the elevated expression of SLC7A11 in selenophilic cancer cells results in increased cystine import, reduction of cystine to cysteine in the cell and export via an undetermined exporter. This results in the extracellular accumulation of thiols (from cysteine), which can reduce selenite to volatile selenide and can induce both its import into the cell and cellular toxicity. It has been known that upregulation of SLC7A11 results in increased glutathione production via cystine import in cancer[19]. The 'moonlighting' role of SLC7A11 and xCT in allowing selenium uptake and selenocysteine biosynthesis provides an elegant mechanism for the coordination of increased glutathione production with increased production of the selenoprotein enzymes (GPXs) that utilize it. Furthermore, this mechanism explains the decades-old observation of increased selenium uptake in cancer cells[13,14] and suggests that selenium uptake has a role in cancer-cell biology, particularly involving the selenoproteins that are produced as a result. SLC7A11 is regulated by crucial cancer genes such as *p53* (ref. [20]), and we speculate that these canonical cancer pathways may thus affect selenocysteine biosynthesis.

Selenoproteins, such as GPX4, have been implicated in the protection of cancer cells against various insults, such as ferroptosis or chemotherapeutic drugs[21,22], so we examined whether selenophilicity is associated with increased resistance to such insults. We found that selenophilic breast-cancer cells had increased resistance against the ferroptosis-inducing prooxidant tert-butyl hydroperoxide (TBH), as well as against nutrient starvation and hypoxia, in contrast to the non-selenophilic normal cells (Fig. 3j,k and Extended Data Fig. 6a,b). We did not note any association of SEPHS2 or GPX4 expression with any tumour environmental features (Extended Data Fig. 7a–f) or with hypoxia in culture (Extended Data Fig. 7g). Knockout of SEPHS2 or GPX4 did not sensitize the selenophilic cancer cells to hypoxia or nutrient starvation, suggesting that other mechanisms downstream of SLC7A11 are involved in their protection against these insults (Extended Data Fig. 6c,d).

In contrast, disruption of selenocysteine biosynthesis via SEPHS2, PSTK or SEPSECS KO, or KO of GPX4 itself, sensitized cells to TBH (Fig. 3i,m), suggesting a role for the selenocysteine biosynthesis pathway in protection against ferroptotic insults in selenophilic cancer cells. Furthermore, the cell death induced by SEPHS2 KO was partially rescued by the ferroptosis inhibitors ferrostatin and liproxatin, suggesting that loss of GPX4 and sensitization to ferroptosis may facilitate the toxicity of SEPHS2 KO (Extended Data Fig. 3e). Recently, FSP1 was identified as an endogenous inhibitor of ferroptosis[23,24]. KO of FSP1 in the MCF10A cells, which were normally resistant to SEPHS2 KO toxicity, rendered them sensitive (Fig. 3n,o). The function of FSP1 explains how these cells can tolerate SEPHS2 KO, even though it results in loss of GPX4. Collectively, these results indicate that upregulation of the selenocysteine pathway is advantageous to cancer cells by protecting against ferroptosis, whereas its loss is tolerable to non-transformed cells owing to the presence of antiferroptotic mechanisms. Furthermore, the toxicity of SEPHS2 KO in cancer cells is facilitated by their hypersensitization to ferroptosis.

SLC7A11-driven selenophilicity provides a protective advantage against ferroptosis through selenoprotein (GPX4) production, but we hypothesized that it will also introduce a requirement for the detoxification of selenium, in particular of the pathway intermediate selenide. Evidencing this double-edged aspect of selenium, increasing doses of selenite supplementation in the selenophilic MDA231 cells resulted in a dose-dependent increase in GPX4 levels (Extended Data Fig. 8a). However, at higher doses, GPX4 expression plateaued, along with the formation of reactive oxygen species (ROS), which is a predicted product of selenide in aqueous solutions[25]. Furthermore, overexpression of catalase (CAT), which decomposes one form of ROS (H_2O_2), partially rescued SEPHS2 toxicity (Extended Data Fig. 8c–e). These results suggest that SEPHS2 and the selenocysteine biosynthesis pathway sequesters free intracellular selenium (such as that in the form of selenide) to form beneficial selenoproteins, but beyond a threshold, the free intracellular selenium not incorporated into selenoprotein can harm cells via ROS formation. Further supporting this notion, selenium delivered via selenocysteine-containing peptides[26] similarly induced GPX4 expression but did not induce ROS formation (Extended Data Fig. 8a,b).

To directly examine whether SLC7A11 makes cells dependent on SEPHS2, we preemptively knocked out or pharmacologically inhibited SLC7A11 prior to SEPHS2 KO (Fig. 4a and Extended Data Fig. 5f–h). We found that not only could cells tolerate SLC7A11 loss, but also this fully prevented the toxicity of subsequent SEPHS2 KO, demonstrating that SLC7A11 directly causes a dependency on SEPHS2, and indicating a gain-of-function mode of toxicity.

To further determine whether SEPHS2 has a function that is independent of its role in selenoprotein synthesis, we took advantage of the fact that KO of the other selenocysteine biosynthesis components PSTK or SEPSECS can also eliminate selenoprotein production, including that of SEPHS2. We constructed a mutant SEPHS2 (SEPHS2-U60C) with a selenocysteine-to-cysteine mutation that is predicted to decrease, but not eliminate, catalytic activity[27] while allowing the protein to be expressed even when selenocysteine biosynthesis is halted owing to SEPSECS or PSTK KO. We demonstrated that this enzyme is functional, as it rescued SEPHS2 KO in our earlier experiment (Fig. 1e). Expression of this construct was found to rescue the toxicity of SEPSECS or PSTK KO despite the cells' inability to express any other selenoprotein (Fig. 4b,c), which demonstrates that (1) SEPSECS and PSTK KO are toxic because of the loss of SEPHS2 that they cause, and (2) SEPHS2 impacts viability even when all selenoprotein biosynthesis has ceased. The rescue was not complete, indicating that loss of other selenoprotein(s) such as GPX4 (ref. [27]) may have partially contributed to the toxicity. Collectively, these data demonstrate that SEPHS2 is required in cancer cells for a reason other than production of selenoproteins.

We next examined whether SEPHS2 is required in cancer cells to detoxify the selenide produced by the selenocysteine biosynthesis pathway. Supporting this model, overexpression of SEPHS2 protected cancer cells against the toxicity of selenite, whereas loss of SEPHS2 created a synergistic effect with selenite (Fig. 4d,e). As selenide is volatile and cannot be detected by conventional liquid chromatography–mass spectrometry (LC–MS) methods[28,29] nor treated as a conventional toxic drug, we had to develop a system for treating selenide to cells in a controlled manner, as well as a method for directly detecting selenide. First, we dissolved sodium selenide (Na_2Se) in an acidic (3.7% HCl) solution, which is predicted to produce selenide (H_2Se) via replacement reaction (Fig. 4f). To detect whether selenide gas was produced, we embedded silver nitrate or lead acetate, which have been shown to react with selenide in aqueous solution to produce dark brown metal–selenium compounds[30], into polyvinylpirolidone (PVP) as an immobilizing matrix, and these spots were placed on the lid of 96-well plates at varying distances from the gas-producing compartment. A colour change was observed that directly correlated with distance, suggesting the capture of volatile selenide in these spots (Fig. 4f and Extended Data Fig. 10f). Importantly, inductively coupled plasma mass spectrometry (ICP-MS) of these spots indicated that selenium had a robust presence, verifying that we were capturing volatile selenide gas (Fig. 4g). When this reaction was conducted in the middle compartment of a 96-well plate confluent in each well with MDAMB231 cells, we observed toxicity that followed a radial pattern emanating from the gas source, with toxicity correlating with proximity (Fig. 4h and Extended Data Fig. 10g). Using this treatment system, we directly tested the effects of disrupting or overexpressing SEPHS2 on cells plated at equal distances from the source of selenide gas (Fig. 4i). Disruption of SEPHS2 sensitized cancer cells to selenide (Fig. 4j), and overexpression of SEPHS2 protected cancer cells against it (Fig. 4k). Therefore, these results clearly demonstrate that SEPHS2 has a selenide-detoxification function.

In summary, we have shown how selenium uptake and selenocysteine biosynthesis are aberrantly activated in a subset of cancer cells owing to SLC7A11/xCT, and how this necessitates selenide detoxification by SEPHS2. The identification of selenophilic subsets of cancer cells has important implications in cancer biology that future investigative efforts can address, as selenium uptake drives the expression of selenoproteins implicated in processes such as ferroptosis, drug resistance and endoplasmic-reticulum stress[1,21,22]. While upregulation of selenoprotein production via the SLC7A11–SEPHS2 axis is likely to impact various aspects of cancer biology, we demonstrate that the most urgent requirement in these cancer cells is the detoxification of selenide, an endogenously produced metabolite that shows high volatility and toxicity. Indeed, our functional demonstration of selenide as the toxic intermediate of selenocysteine biosynthesis explains the longstanding paradox of how selenium, an essential micronutrient, can be toxic and anticarcinogenic. Targeting the selenide detoxifier SEPHS2 allows

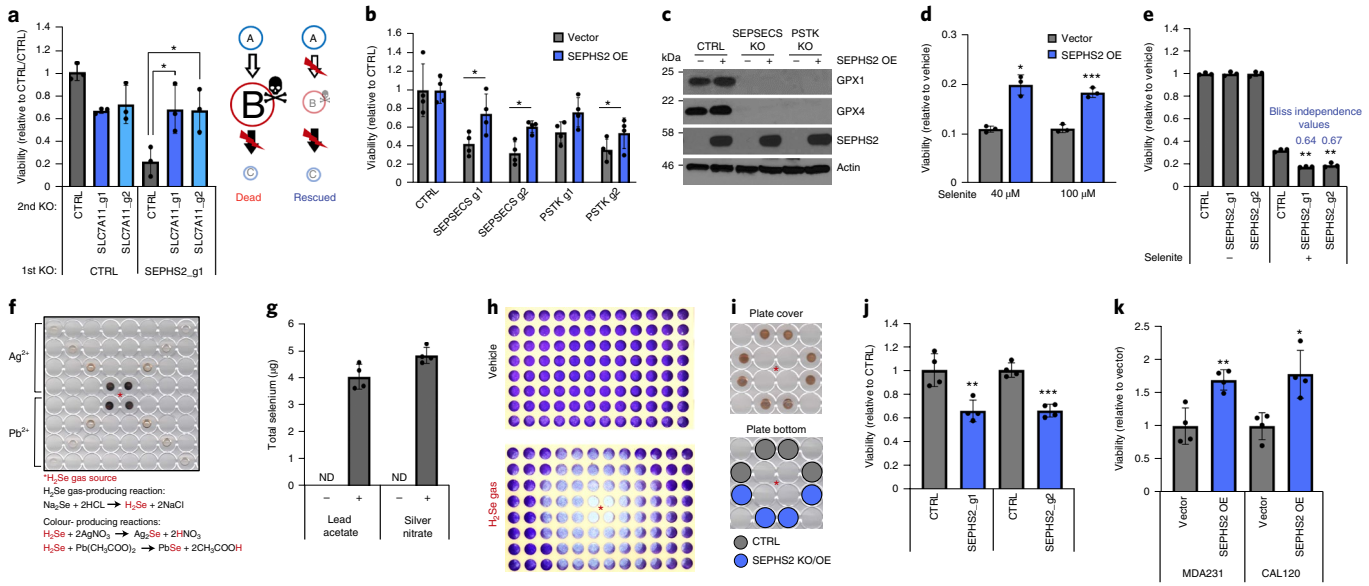


Fig. 4 | Essentiality of SEPHS2 in cancer cells is due to its role as a selenide-detoxifying enzyme. a, Viability of MDAMB231 cells subjected to sequential gene KO as shown, demonstrating that preemptive loss of SLC7A11 renders SEPHS2 dispensable. Values were normalized to those of cells from the same line subjected to CTRL KO and then subjected to CTRL KO. **b**, Viability of U251 cells overexpressing (OE) blank vector or SEPHS2-U60C that were subjected to KO with gRNAs against *SEPSECS* or *PSTK* for 11 d. **c**, Immunoblot of selenoproteins in U251 cells overexpressing blank vector or SEPHS2-U60C and subjected to KO with gRNAs against *SEPSECS* or *PSTK* for 11 d. **d**, Viability of U251 cells overexpressing blank vector or SEPHS2-U60C and treated with sodium selenite for 48 h. Values are relative to that of cells from the same line treated with vehicle. **e**, Viability of U251 cells subjected to KO with gRNAs against *SEPHS2* or controls and then treated with 12 μM sodium selenite for 72 h. Bliss independence values were calculated with the formula $E_C = E_A + E_B - E_A \times E_B$, for which a value below 1.0 indicates a synergistic effect. **f**, Scanned image of the polystyrene microplate cover spotted with PVP embedded with silver nitrate or lead acetate, used as a non-contact selenide-gas indicator, after selenide-gas exposure for 1 min. **g**, Selenium quantification, measured by ICP-MS, of the spots of PVP embedded with silver nitrate or lead acetate, demonstrating the capture of volatile selenide. Four spots that were equidistant from the selenide source were measured. ND, not determined. **h**, Scanned image of crystal violet staining of the remaining MDAMB231 cells following vehicle or selenide-gas exposure for 24 h. For **f** and **h**, the asterisk indicates the location of selenide-gas production. **i**, Experimental layout for comparing sensitivity to selenide gas, via equidistant cell plating. Similar browning of silver nitrate–PVP spots indicates similar exposure levels to selenide gas. **j**, Viability of SEPHS2-KO or control MDAMB231 cells under selenide-gas exposure for 24 h. Values are normalized to the viability of cells transduced with non-targeting gRNAs. **k**, Viability of MDAMB231 cells overexpressing SEPHS-U60C or blank vector under selenide-gas exposure for 48 h. Values are normalized to the viability of empty-vector-transduced cells. For **a** and **b** and **d** and **e**, $n = 3$ biological replicates; for **j** and **k**, $n = 4$ biological replicates; for **g**, $n = 4$ technical replicates; error bars are s.d. For all panels, the measure of centre is the mean. For **a** and **c**–**k**, $^*P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ (two-tailed Student's *t*-test). For **b**, $^*P < 0.05$ (two-tailed paired Student's *t*-test).

us to kill cancer cells with efficacy and precise selectivity, serving as a blueprint for future studies to exploit other pathways containing toxic metabolites.

Methods

Materials used. Details for all chemicals, DNA constructs, antibodies and other materials are provided in Supplementary Table 3. Details for all cell lines are provided in Supplementary Table 2.

Cell lines and cell culture. All cell lines were cultured at 37 °C under 5% CO_2 and atmospheric oxygen. The cell lines used in this paper are listed in Supplementary Table 2 along with information on their media and sources. Short tandem repeat (STR)-based cell-line verification information is also provided in Supplementary Table 2.

Molecular cloning of expression constructs. SEPHS2 was PCR-amplified from complementary DNA produced from WM115 cells, using forward and reverse primers containing BamH1 and NOT1 sites, respectively. The verified amplicon sequence corresponded exactly to the published consensus sequence for SEPHS2 (CDS ID: CCDS42150.1). The amplicon was cloned into pLV-EF1α-IRES-Blast vector. SEPHS2-U60C was constructed with a region (c.180A>C for p.U60C, and 5′-GGAGGGACGGCAGTGACCGGTGG-3′ (616–638 bp) was replaced by 5′-GGAGGGACCGCCGTTACTGGTGG-3′ for SEPHS2 CRISPR g1 resistance) mutated by site-directed mutagenesis for CRISPR resistant SEPHS2-U60C. Wild-type SEPHS2 has a UGA codon (amino-acid position 60), which acts to encode selenocysteine due to the presence of a stem-loop structure in the 3′ untranslated region called the selenocysteine insertion sequence (SECIS). In our SEPHS2-U60C construct, the codon is mutated to cysteine and the SECIS removed to allow expression independently of selenocysteinyl-tRNA availability, and four silent mutations are present in the guide-recognition site for SEPHS2_g1. All primer sequences are described in Supplementary Table 1.

Lentivirus production and CRISPR–Cas9-mediated genome editing in cell lines. Optimal gRNA sequences were obtained from a published guide sequence library[31]. Pseudotyped lentivirus was produced by cotransfecting pLentiCRISPR v2-containing guide sequences of interest[7] with the VSV-G envelope plasmid and the Delta-Vpr packaging plasmids into HEK293T cells using X-tremeGENE 9 Transfection Reagent (Roche). Lentivirus-containing supernatant was collected at 24 h after replacement with fresh medium. The virus was directly used without being concentrated. Target cells were infected with freeze and thawed viral supernatant in the presence of 10 μg ml^{-1} polybrene for 24 h and then replaced in medium with puromycin (typically at 1 μg ml^{-1}) or blasticidin (typically, 20 μg ml^{-1}) to select for transduced cells. All guide sequences are listed in Supplementary Table 1. The titre for each virus was determined, and cells were infected at sub-one multiplicity of infection to prevent non-specific toxicity that can occur from superinfection leading to excessive expression of Cas9 and guide.

Determining relative cell viability following CRISPR–Cas9 KO. Varying numbers of cells for each cell line (between 1,000 and 5,000 cells per well in each 96-well plate) were plated, and 1 d after plating, cells were infected with lentivirus and polybrene. As the cell lines have differing growth rates, infectivity to pseudotyped lentivirus and sensitivity to puromycin, each of these parameters had to be optimized for each cell line. Briefly, the goal was to adjust the starting number of cells and the amount of virus for each cell line such that, after 3 d of puromycin selection, roughly 100–200 infected cells survived selection and were visible in the well. This low number ensured a low multiplicity of infection, and that cells did not reach confluency before the end of the experiment, which would introduce noise.

For each CRISPR guide–cell line combination, we measured cell viability (using CellTiter-Glo) at two points following infection and selection (day 5 and day 9).

For each combination, the relative growth, as measured by fold-change increase of CellTiter-Glo signal between these two time points, was calculated and compared with the growth rate of that cell line transduced with non-targeting (CTRL) guide. For example, if U2OS cells grew by 5-fold under CTRL gRNA between day 4 and day 9 and U2OS cells grew by 2.5-fold under guide X, then guide X would have a relative viability score of 0.5. This system allowed us to precisely control for both the different growth rates and infectivity of different cell lines, as well as variabilities in starting number of infected cells, which arise even with titered virus. In this manner, the effects of KO for each of the 12 candidate detoxifying enzymes, 3 guides each, were determined across 11 different cells lines for the initial analyses (Fig. 1b). As we expanded to examine SEPHS2 KO effect in more cell lines (Fig. 1c), we utilized two guides (SEPHS2_g1 and SEPHS_g2). The individual guides are averaged to represent SEPHS2 effect.

Monitoring of long-term growth of MDAMB231 and MCF10A CRISPR–Cas9 KO lines. MDAMB231 and MCF10A cells were infected with lentivirus to transduce them with CRISPR–Cas9 gRNAs against CTRL or SEPHS2 (guide 1), each in a 10-cm dish. The day after infection was counted as day 1. Every 4 d, cell counts were obtained and recorded, and 500,000 cells were replated onto a new 10-cm dish, to have sufficient cells to continue the experiment and to allow monitoring of the growth rate of these 500,000 cells over the next 4 d. Leftover cells that were not replated were lysed for western blot analyses as shown in Fig. 2d. This step was repeated every 4 d to continually monitor the growth rate of each cell line.

Colony-formation assay. MDAMB231 cells expressing SEPHS2-U60C or blank vector, and MCF10A cells were infected with SEPHS2 KO or CTRL virus. After 6 d of selection, the cells were seeded at a density of 2,000 MDAMB231 cells or 500 MCF10A cells per well in 6-well plates with 2 ml medium; MCF10A cells were plated at a lower density because of their higher proliferation rate. Five hundred microlitres of medium was added in each well every 4 d without medium change. Nine days after the initial cell seeding, cells were washed with PBS, and stained for 1 h with 0.5% crystal violet in 25% methanol/75% PBS solution. Plates were then washed twice with PBS to remove excess dye and were scanned.

Western blot analysis. Briefly, cells were quickly washed with cold PBS and lysed in cold RIPA lysis buffer freshly supplemented with protease inhibitor (Roche). Supernatant was collected after centrifugation at 13,000*g* at 4 °C for 10 min. Protein concentration was determined by Bradford assay. Proteins were denatured in 6× Laemmli buffer (Boston Bioproducts) and boiled at 90 °C for 5 min. We found that not boiling the sample was required for SLC7A11 detection. Equal amounts of total protein (typically, 15–25 µg depending on the experiment) were loaded per lane on a polyacrylamide gel and analysed by standard immunoblotting. Primary antibody dilutions are provided in the Reporting Summary. Immunoblot signals were detected by ECL, Pico or diluted Femto substrate (Pierce).

Cell-viability assay. Cell viability was measured by CellTiter-Glo Luminescent Assay (Promega) using a Beckman Coulter DTX880 Multimode plate reader. Luminescence capture was set at 1 s. Crystal violet staining was performed to visualize surviving cells following exposure to hydrogen-selenide gas in a 96-well plate (Fig. 4h). Briefly, 8,000 cells were plated in two 96-well plates. The next day, the reaction mixture, which produced hydrogen-selenide gas, was placed in the centre of the plate, and the plate was tightly sealed in the plastic bag. The plate sealed in the plastic bag was placed in the CO_2 incubator. After 24 h of incubation, cells were washed with PBS, and stained for 1 h with 0.5% crystal violet in 25% methanol/75% PBS solution. Plates were then washed twice with PBS to remove excessive dye and scanned.

Medium thiol quantification. Cells were washed once with phenol-red-free growth medium, and 10,000 cells were plated with the 100 µl of phenol-red-free growth medium in a 96-well plate. One hundred microlitres of conditioned medium was collected and directly mixed with 50 µl of 10 mM DTNB (5,5′-dithiobis-(2-nitrobenzoic acid)) dissolved in DMSO after 24 h of conditioning. Absorbance at 450 nm was measured spectrophotometrically in 5 min (DTX880, Beckman Coulter). The remaining cells in the same plate were subjected to CellTiter-Glo assay for normalization. The absorbance value of the blank was subtracted as background, and values were normalized to total viable cells as determined by CellTiter-Glo. Owing to the masking effect of phenol red on DTNB absorbance, created by reacting with thiols, phenol-red-free medium was used for all thiol quantifications.

Production of selenide. Selenide gas was formed by mixing sodium selenide, at a final 50 mM concentration, into 3.7% hydrochloric acid solution. Because selenide gas is volatile, 100 µl of the mixture was added into the centre of the 96-well plate immediately after mixing the sodium selenide with hydrochloride. Immediately following this, the plate was tightly sealed in the plastic bag. All procedures were performed in the safety hood because of the toxic and volatile nature of the selenide that was produced.

Selenide detection. PVP (5% wt/vol) solution was mixed with 1 M silver nitrate or lead acetate at a ratio of 9:1, yielding final concentrations of 100 mM of silver nitrate or lead acetate, and 4.5% PVP. Reagents were freshly prepared for each experiment. The solutions were mixed well by pipetting. Next, 15 µl of metal–PVP mixture was carefully spotted on the inside of the cover of a 96-well plate. The cover was dried at room temperature for at least 1 h, to form a coating membrane. This cover containing the 'detector spots' was placed above the wells, in which selenide gas was produced in the centre of the plate. Upon immediate reaction with the selenide gas, the spots produced silver selenide or lead selenide, which are brown in colour whereas the original spot was clear. Spots were developed for up to 1 min. The plate was imaged, and the spots were subjected to acid digestion for further ICP-MS analysis.

Total selenium measurement by ICP-MS analysis. Cells were plated into 15-cm dishes, and medium was changed at 75% confluency and treated with 12 µM selenite 24 h after medium change. Cells were collected at 2 h after selenite treatment, washed 3 times in cold PBS and weighed to a total cellular mass of 80–100 mg. The cell pellets were treated with 500 µl 1:1 H_2O_2/HNO_3 and were allowed to stand for 12 h at room temperature with periodic venting. The samples were then sonicated at 35 kHz and 40 °C for 1 h with periodic venting. The contents of the Eppendorf tube were transferred to a glass digestion tube with ASTM type I water (2 × 1 ml). The tubes were sealed and heated at 140 °C for 2 h, cooled and diluted to 5 ml with ASTM type I water for analysis.

All analyses were carried out with an Agilent 7500 A ICP-MS system fitted with a standard concentric nebulizer, a Peltier-cooled double-pass Scott-type spray chamber, a torch shield and standard Ni interface cones. The peristaltic pump was set to deliver 10 g l^{-1} yttrium as an internal standard, and masses at 77–78, 82, 83 and 89 were scanned using a spectrum analysis with 20 points per mass, with 3 repetitions. Interference correction was conducted on the mass at 82 AMU using standard environmental correction for bromine (82 = 82–83). Following each analytical sample, the probe-sample introduction system was rinsed with 10% nitric acid for 1 min to minimize carry-over. Calibration curves were prepared using standard solutions of Se (Ultra Scientific) and quality-control samples using a multielement standard from Environmental Express. In addition, mock samples were prepared for each calibration level and subjected to the sample preparation described previously to assure adequate recovery. In all cases, the recoveries were 100 ± 10% (data not shown). The calibration solutions were analysed without background subtraction using the 89 internal standard peak. In all cases, the linear regression afforded an $r \geq 0.996$. The relative concentrations based on the use of the 77 and 82 isotopes were examined and found to agree within ±10%, so the average concentration was reported.

Measurement of selenium uptake by fluorometric assay. Total selenium content was measured via fluorometric assay[32]. Cells were plated into 10-cm dishes, medium was changed at 75% confluency and was treated with vehicle or 12 µM selenite at 24 h after medium change. Cells were collected 2 h after selenite treatment, washed three times in cold PBS, and counted to 5 million then resuspended in 100 µl MiliQ H_2O and added to borosilicate glass tubes. Then, 500 µl of a 4:1 mixture of HNO_3 and $HClO_4$ was added to the glass tube and incubated on a heat block overnight at 130 °C. Next, 500 µl of HCl was then added and incubated at 130 °C for 1 h. Tubes were removed and allowed to cool to room temperature before addition of 2 ml of 2.5 mM ETDA, 500 µl of 6.3 mM 2,3-diaminonapthalene-HCl in ultrapure water and 1 ml of cyclohexane. Tubes were capped and incubated for 30 min at 60 °C. Samples were placed in a shaker for 5 min, and then 150 µl of the upper phase was aliquoted in technical triplicates into a 96-well plate. The plate was read for fluorescence emission on a DTX880 (Beckman Coulter) plate reader with excitation at 360 nm and emission at 528 nm. Values for selenite-supplemented cells were normalized to basal values of intracellular selenium, giving a relative selenium-uptake measurement.

Orthotopic xenograft studies. Eight-week-old female wild-type nude athymic nu^-/nu^- mice were purchased from Charles River Laboratories and raised in an animal facility at University of Massachusetts Medical School (UMMS). Littermate controls were not used. To compare tumour growth between control and SEPHS2-KO cell lines in vivo, at 10 d after viral transduction, the cells (1×10^6) were resuspended in 1× PBS with Matrigel basement membrane matrix at a 1:1 ratio (total volume, 35 µl) and were subcutaneously injected into the area between the upper right second and third mammary fat pads of the nude mice. Tumour sizes were measured using a Vernier caliper, and volumes were determined according to the formula $4/3\pi \times ((L \times W \times D)/2)$, where L is the measurement parallel to the midline of the supine animal and W is perpendicular to the midline, and D is the distance from chest to the top of the tumour. Mice were euthanized at 7 weeks after injection, and the solid tumours were isolated. Mouse orthotopic xenografts were set up with 7 mice per cohort, as we have previously obtained statistically significant results with this number of mice using this cell-line model in previous experiments that projected a >50% difference in cell viability. Mice were randomly assigned to be injected with SEPHS2 KO MDAMB231 cells or CTRL KO MDAMB231 cells. The researchers who carried out injection and tumour measurements were blinded to groups (SEPHS2 KO versus CTRL KO tumours). Our experiments complied with all ethical regulations and were approved by the Institute Animal Care and Use Committee (IACUC) of University of Massachusetts Medical School.

Processing of human breast tissues for SEPHS2 protein quantification. All human breast-cancer samples and normal breast tissues were obtained with informed consent from the University of Massachusetts Medical School Biorepository and Tissue Bank using procedures conducted under an Institutional Review Board (IRB)-approved protocol. After their surgical removal, fresh tumours and normal tissues were partly snap-frozen in liquid nitrogen immediately and stored at −80 °C. The frozen section was homogenized in RIPA buffer with complete protease inhibitor cocktail, and was then centrifuged at 13,000*g* at 4 °C for 10 min. Then SDS–polyacrylamide gel electrophoresis was performed as described in the 'Western blot analysis' section. A strong non-specific band that appeared in breast tissues required the use of a 10% gel and long electrophoresis to separate SEPHS2 protein from the non-specific band (labelled with an asterisk in Fig. 1h). Protein levels of SEPHS2 and actin were quantified using ImageJ. Scanned films were inverted, and the intensity of each band was measured, and then the background value was subtracted. The intensity value of the SEPHS2 protein band for each sample was normalized to the intensity value for actin protein.

Survival analysis. Kaplan–Meier curves was used to analyse tumour-free survival in mouse experiments and for overall survival of human participants. The R survival package[33,34] was used to generate Kaplan–Meier survival plots and to compare the survival distributions of people with low SEPHS2 and those with high SEPHS2. The survival data 'brca_tcga_pub2015_all'[35] was downloaded from cBioportal[36]. People were divided into high and low groups on the basis of their SEPHS2 expression level (top 25% or bottom 25% in the cohort).

Transporter CRISPR screen. We infected U251 cells with two CRISPR guides for each transporter target. Cells were replated in 12 μM selenite medium after full selection of the CRISPR guides, typically for 5 d. We measured cell viability (using CellTiter-Glo) at 3 points post selenite treatment: 24, 48 and 72 h. For each guide, the relative growth was measured by fold-change increase of CellTiter-Glo signal between selenite-treated cells and untreated controls.

Double-KO rescue assay. MDAMB231 cells were infected with pLentiCRISPR V2-based lentivirus to deliver the first KO (CTRL, SLC7A11_g1, SLC7A11_g2) and were selected with puromycin. Cells were secondarily infected with pMD154 lentivirus to allow expression of the second gRNAs (CTRL or SEPHS2_g1) 6–7 d after initial infection, and were subsequently selected with 500 μg ml^{-1} hygromycin. These cells were selected for 5 d before being plated in triplicate in 96-well plates, with 200 cells per well and with duplicate plates. One plate was assayed for viability after overnight incubation, and the second plate was read after 5 d of growth to determine the relative growth rate of double-KO cells in comparison to controls.

Immunohistochemistry. Immunohistochemical detection of CD31, Ki67, CA9 and GPX4 in serial tumour samples was performed using formalin-fixed and paraffin-embedded tissue sections. After deparaffinization in xylene for 20 min, rehydrated tissue sections were subjected to antigen retrieval by boiling in Borg Decloaker solution (Biocare Medical) for 5 min. Slides were blocked with 5% horse serum and then incubated with antibodies for 1 h at room temperature. After incubation, biotinylated secondary antibodies plus streptavidin were used to stain the samples, with DAB chromogen as substrate, and haematoxylin solution for nuclear counterstaining. The section samples were imaged by QCapture Suite 2.98.2.

Immunocytochemistry. Cell were seeded on cover slips coated with poly-D-lysine and collagen, fixed with 4% paraformaldehyde in PBS, permeabilized by incubation with 0.1% Triton X-100 in PBS and subsequently blocked with 1% BSA in PBS for 1 h. To quantify the level of cells expressing cleaved ecaspase-3, the cells were incubated with a rabbit antibody against cleaved caspase-3 (Asp175) (Cell Signaling, 9664) in the cold room overnight. The cells were washed with PBS 3 times and were incubated with anti-rabbit Alexa Fluor 488 (Invitrogen) for 1 h at room temperature in the dark. The cells were subsequently washed with PBS, and mounted with Gold Antifade Reagent with DAPI (Invitrogen, P36935).

RNAscope in situ hybridization. For SEPHS2 mRNA staining, fluorescent RNAscope in situ hybridization (ISH) was performed on the paraffin-embedded serial tumour sections using the RNAscope protocol (Advanced Cell Diagnostics, RNAscope Multiplex Fluorescent Reagent Kit v2 Assay User Manual, document number; 323100-USM); all reagents were provided by the RNAscope fluorescent reagent kit (Supplementary Table 3). After deparaffinization in xylene for 20 min and rehydration with serial 100%, 95% and 70% ethanol for 5 min each, tissue sections were treated with hydrogen peroxide for 10 min and heated at 99 °C in target-retrieval solution for 20 min. After target retrieval, sections were treated with protease plus at 40 °C in the hybridization oven for 30 min and hybridized with 3-plex positive control RNA probe, 3-plex negative control RNA probe and SEPHS2 RNA probe for 2 h at 40 °C in the oven. The section samples were incubated with 3 different amplifiers, AMP1 (30 min), AMP2 (30 min) and AMP3 (15 min), at 40 °C to amplify hybridization signals. After amplification steps, sections were treated with horse radish peroxidase C1 for 15 min at 40 °C and incubated with diluted TSA plus fluorescein (1:50 in TSA buffer) for 30 min at 40 °C in the oven. The section samples were counterstained and mounted with Vectashield antifade mounting medium with DAPI and were imaged by Lionheart LX Automated microscope (BioTek). Images were processed by Gen5 software (BioTek).

Cell-death analysis. Cell death was measured using a FITC Annexin V Detection Kit (BD Biosciences, 556547). Cell lines treated with selected concentrations of sodium selenite or cell lines following KO with guides against SEPHS2 were stained with propodium iodide and annexin V–FITC. The stained cells were analysed using the BD LSR II flow cytometer (BD Biosciences). The data were analysed by FlowJo 10. The gating and quadrant strategy for FACS measurement is provided in Extended Data Fig. 10.

Detection of ROS. ROS were detected using DCFDA Cellular ROS Detection Assay Kit (Abcam, ab113851). MDAMB231 cells were seeded at 2,000 cells per well, and 24 h later were stained with DCF reagent and then treated with various selenite concentrations for 45 min. For SeCys peptide treatment, MDAMB231 cells were seeded at 2,000 cells per well and, 24 h later, were treated with various concentrations of SeCys peptide for 24 h before DCF staining. Fluorescence was measured using 485-nm excitation and 535-nm emission filters with a Beckman Coulter DTX880 Multimode plate reader. Values were normalized to those of DCF background and vehicle treatment.

Medium conditioning, selenide uptake and ferroptosis. We have established that selenite uptake is highly reliant on the presence of extracellular thiols (Fig. 3d–i). Therefore, we observe that the toxicity of selenite treatment is highly modulated by whether the medium has been preconditioned for a period of time, which increases toxicity, or whether selenite has been added along with a fresh medium change, which conversely requires a longer time for selenium uptake and toxicity to occur. We note that in the experiments for Figs. 3b,d and 4d,e and Extended Data Fig. 5f,g, selenite was added to the medium that had already been conditioned since the seeding of the cells; in Fig. 3a and Extended Data Fig. 8a, selenite was added along with fresh medium. Owing to the need to prestain cells for the DCFDA assay in Extended Data Fig. 8b, selenite was added with 75 μM reduced glutathione in assay buffer to provide reduced thiols because conditioned medium could not be used for this assay.

We found that a second factor that affected toxicity in some of our experiments was whether the virally transduced cells were 'in place' for the duration of the experiment, or for practical reasons if trypsin detached and then reattached during the experiment, for example when cells were virally transduced and selected in 6-well format and then replated in 96-well format for CellTiter-Glo assays. In the latter case, we found that disruption of selenocysteine biosynthesis pathway (such as via SEPHS2 KO) did induce a small degree of toxicity even in the insensitive cells, which we think was due to detachment and reattachment being a mild ferroptotic stimulus. The MCF10A cells from Fig. 3n and Extended Data Fig. 2c show this effect, compared with the same cells in Fig. 1d, which were not detached and reattached.

Statistics and reproducibility. Tumour-free survival in mice was compared between CTRL and SEPHS2-KO groups using the Kaplan–Meier method, and significant differences in curves were assessed using the log-rank test. Results of the viability assay and Ellman's test and total selenium measurement were analysed using a two-tailed Student's *t*-test. $P < 0.05$ was considered statistically significant, and data marked with one (*), two (**) or three (***) asterisks indicate *P* values of < 0.05, < 0.01 and < 0.001, respectively. All experiments were repeated at least three times with the following exceptions: focused candidate detoxifying-enzyme screen was carried out in 11 cell lines in Fig. 1b, effects of SEPHS2 KO on tumour growth (Fig. 1f,g and Extended Data Fig. 2f,g), focused screen for selenium=uptake mediator (Fig. 3a), SEPHS2 protein measurements in tumours versus normal tissues (Fig. 1h,i), imaging for cleaved caspase-3 in SEPHS2 KO cells (Extended Data Fig. 3c,d), selenium ICP-MS quantification of captured spots (Fig. 4g), validation of SEPHS2 expression KO (Extended Data Fig. 9) and validations of drug efficacy (Fig. 3j and Extended Data Fig. 8c), which were performed once. The following experiments were repeated twice: SEPHS2 KO in multiple cell lines (Fig. 1c, d), SEPHS1 KO in multiple cell lines (Extended Data Fig. 2i), SEPHS2/GPX4/SEPSECS KO cell treatment with hypoxia and nutrient starvation (Extended Data Fig. 6c,d) and erastin treatment of SEPHS2 KO (Extended Data Fig. 5h). In all experiments, all attempts at replication were successful.

Bliss independence values (Fig. 4e) were calculated using standard formula of $E_C = E_A + E_B - E_A \times E_B$. E_A is the effect of SEPHS2 KO alone, E_B is the effect of selenite treatment alone, and E_C is the expected theoretical combination of SEPHS2 KO and selenite treatment. A Bliss independence score of <1 indicates a synergistic effect.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

All raw data that support the findings of this study are available from the corresponding author upon request. Source data are provided with this paper.

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References

1. Labunskyy, V. M., Hatfield, D. L. & Gladyshev, V. N. Selenoproteins: molecular pathways and physiological roles. *Physiol. Rev.* **94**, 739–777 (2014).
2. Burk, R. F. & Hill, K. E. Regulation of selenium metabolism and transport. *Annu. Rev. Nutr.* **35**, 109–134 (2015).
3. Hanahan, D. & Weinberg, R. Hallmarks of cancer: the next generation. *Cell* **144**, 646–674 (2011).
4. Amoedo, N. D., Obre, E. & Rossignol, R. Drug discovery strategies in the field of tumor energy metabolism: limitations by metabolic flexibility and metabolic resistance to chemotherapy. *Biochim. Biophys. Acta Bioenerg.* **1858**, 674–685 (2017).
5. Wexler, P. TOXNET: an evolving web resource for toxicology and environmental health information. *Toxicology* **157**, 3–10 (2001).
6. Kanehisa, M. & Goto, S. KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* **28**, 27–30 (2000).
7. Sanjana, N. E., Shalem, O. & Zhang, F. Improved vectors and genome-wide libraries for CRISPR screening. *Nat. Methods* **11**, 783–784 (2014).
8. Banerjee, B. D., Dwivedi, S. & Singh, S. Acute hydrogen selenide gas poisoning admissions in one of the hospitals in Delhi, India: case report. *Hum. Exp. Toxicol.* **16**, 276–278 (1997).
9. Schecter, A., Shanske, W., Stenzler, A., Quintilian, H. & Steinberg, H. Acute hydrogen selenide inhalation. *Chest* **77**, 554–555 (1980).
10. Strickland, L. B., Dawson, P. J., Santner, S. J. & Miller, F. R. Progression of premalignant MCF10AT generates heterogeneous malignant variants with characteristic histologic types and immunohistochemical markers. *Breast Cancer Res. Treat* **64**, 235–240 (2000).
11. Xu, X. M. et al. Selenophosphate synthetase 2 is essential for selenoprotein biosynthesis. *Biochem. J.* **404**, 115–120 (2007).
12. Noinaj, N. et al. Structural insights into the catalytic mechanism of Escherichia coli selenophosphate synthetase. *J. Bacteriol.* **194**, 499–508 (2012).
13. Cavalieri, R. R., Scott, K. G. & Sairenji, E. Selenite (75Se) as a tumor-localizing agent in man. *J. Nucl. Med.* **7**, 197–208 (1966).
14. Cavalieri, R. R. & Scott, K. G. Sodium selenite Se 75. A more specific agent for scanning tumors. *JAMA* **206**, 591–595 (1968).
15. Jiang, C., Wang, Z., Ganther, H. & Lu, J. Distinct effects of methylseleninic acid versus selenite on apoptosis, cell cycle, and protein kinase pathways in DU145 human prostate cancer cells. *Mol. Cancer Ther.* **1**, 1059–1066 (2002).
16. Sato, H. et al. Redox imbalance in cystine/glutamate transporter-deficient mice. *J. Biol. Chem.* **280**, 37423–37429 (2005).
17. Sato, H., Tamba, M., Ishii, T. & Bannai, S. Cloning and expression of a plasma membrane cystine/glutamate exchange transporter composed of two distinct proteins. *J. Biol. Chem.* **274**, 11455–11458 (1999).
18. Olm, E. et al. Extracellular thiol-assisted selenium uptake dependent on the x(c)- cystine transporter explains the cancer-specific cytotoxicity of selenite. *Proc. Natl Acad. Sci. USA* **106**, 11400–11405 (2009).
19. Kim, J. Y. et al. Human cystine/glutamate transporter: cDNA cloning and upregulation by oxidative stress in glioma cells. *Biochim. Biophys. Acta* **1512**, 335–344 (2001).
20. Jiang, L. et al. Ferroptosis as a p53-mediated activity during tumour suppression. *Nature* **520**, 57–62 (2015).
21. Hangauer, M. J. et al. Drug-tolerant persister cancer cells are vulnerable to GPX4 inhibition. *Nature* **551**, 247–250 (2017).
22. Yang, W. S. et al. Regulation of ferroptotic cancer cell death by GPX4. *Cell* **156**, 317–331 (2014).
23. Doll, S. et al. FSP1 is a glutathione-independent ferroptosis suppressor. *Nature* **575**, 693–698 (2019).
24. Bersuker, K. et al. The CoQ oxidoreductase FSP1 acts parallel to GPX4 to inhibit ferroptosis. *Nature* **575**, 688–692 (2019).
25. Spallholz, J. E. On the nature of selenium toxicity and carcinostatic activity. *Free Radic. Biol. Med.* **17**, 45–64 (1994).
26. Alim, I. et al. Selenium drives a transcriptional adaptive program to block ferroptosis and treat stroke. *Cell* **177**, 1262–1279 e1225 (2019).
27. Ingold, I. et al. Selenium utilization by GPX4 is required to prevent hydroperoxide-induced ferroptosis. *Cell* **172**, 409–422 e421 (2018).
28. Delafiori, J., Ring, G. & Furey, A. Clinical applications of HPLC-ICP-MS element speciation: A review. *Talanta* **153**, 306–331 (2016).
29. Suzuki, K. T. & Ogra, Y. Metabolic pathway for selenium in the body: speciation by HPLC-ICP MS with enriched Se. *Food Addit. Contam.* **19**, 974–983 (2002).
30. Dauplais, M., Lazard, M., Blanquet, S. & Plateau, P. Neutralization by metal ions of the toxicity of sodium selenide. *PLoS One* **8**, e54353 (2013).
31. Wang, T., Wei, J. J., Sabatini, D. M. & Lander, E. S. Genetic screens in human cells using the CRISPR–Cas9 system. *Science* **343**, 80–84 (2014).
32. Sheehan, T. M. & Gao, M. Simplified fluorometric assay of total selenium in plasma and urine. *Clin. Chem.* **36**, 2124–2126 (1990).
33. A Package for Survival Analysis in S. Version 2.38 (2015); http//cran.r-project.org/package=survival
34. Harrington, D. P. & Fleming, T. R. A class of rank test procedures for censored survival data. *Biometrika* **69**, 553–566 (1982).
35. Ciriello, G. et al. Comprehensive molecular portraits of invasive lobular breast. *Cancer Cell* **163**, 506–519 (2015).
36. Cerami, E. et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov.* **2**, 401–404 (2012).

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Author contributions

A.E.C. and D.K. conceived the project; A.E.C., N.L. and D.K. designed the research. A.E.C. and N.L. performed most of the experiments with assistance from A.N.M.-O., M.E.S., M.B.D., D.Y., A.B.J. and A.P. In situ hybridization and immunohistochemistry experiments were carried out by S.J.P., N.L. and P.L.G. ICP-MS-based experiments for selenite measurements were conducted by M.S. R.L., N.L., M.B.D. and L.J.Z. conducted mining and analyses of TCGA and other large-scale datasets. A.N.M.-O. and L.M.S. assisted with mouse xenograft studies. K.S. assisted with human sample studies. A.E.C., N.L. and D.K. wrote the manuscript with consultation from all of the authors.

Competing interests

A.E.C., N.L. and D.K. are listed as the authors on a patent application filed by University of Massachusetts Medical School on targeting SEPHS2 in cancer therapy, and on the methodology for hydrogen-selenide detection. All other authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s42255-020-0224-7.

Supplementary information is available for this paper at https://doi.org/10.1038/s42255-020-0224-7.

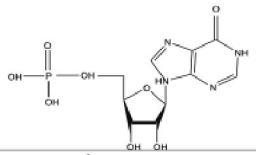
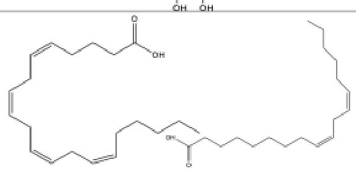
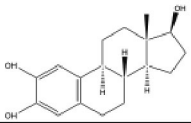
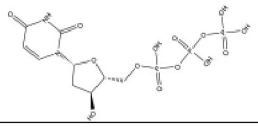
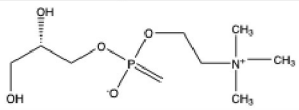
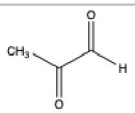
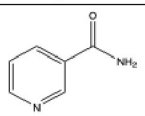
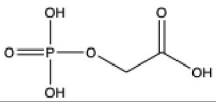
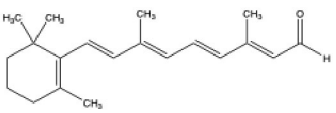
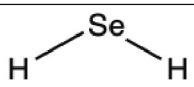
Correspondence and requests for materials should be addressed to D.K.

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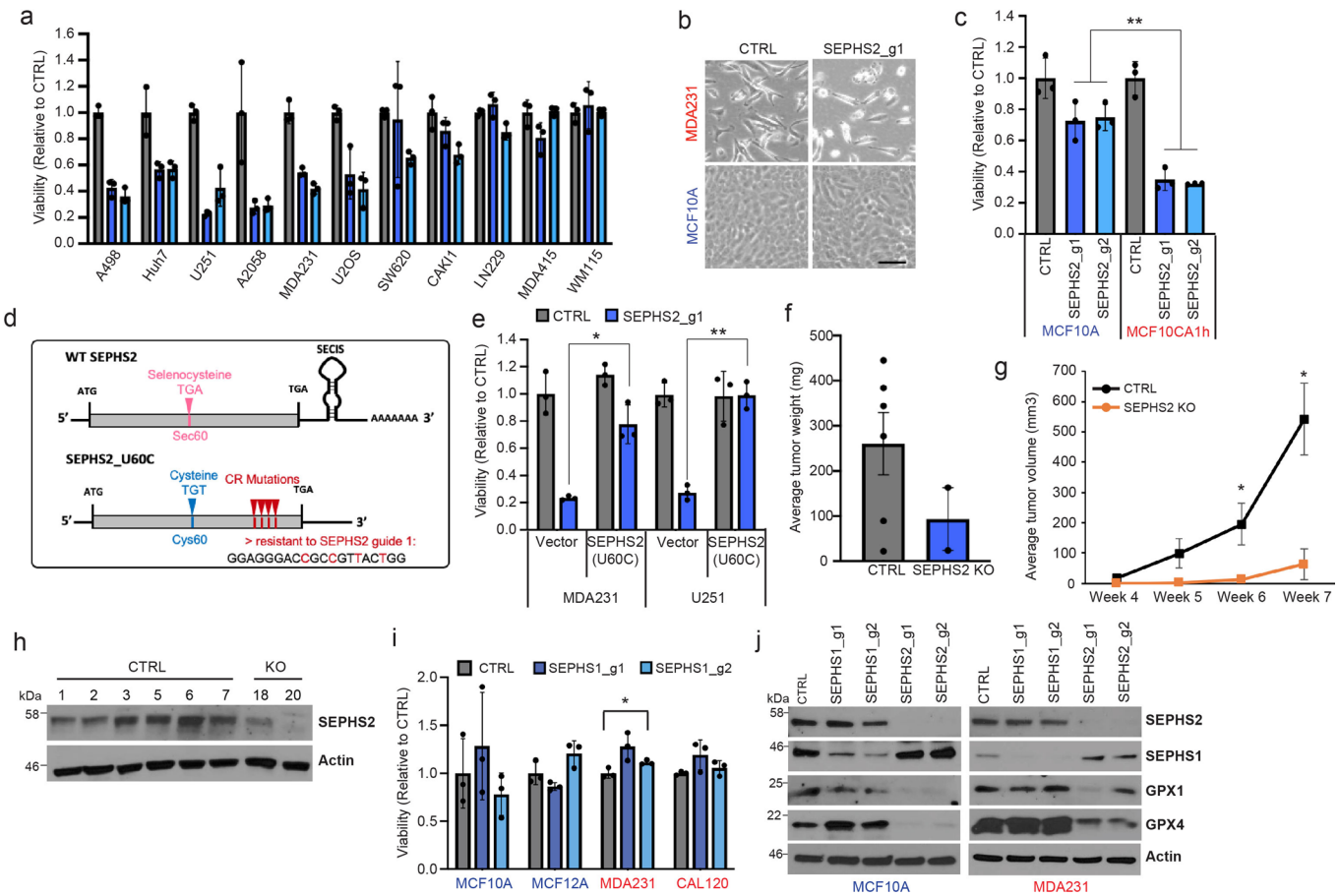
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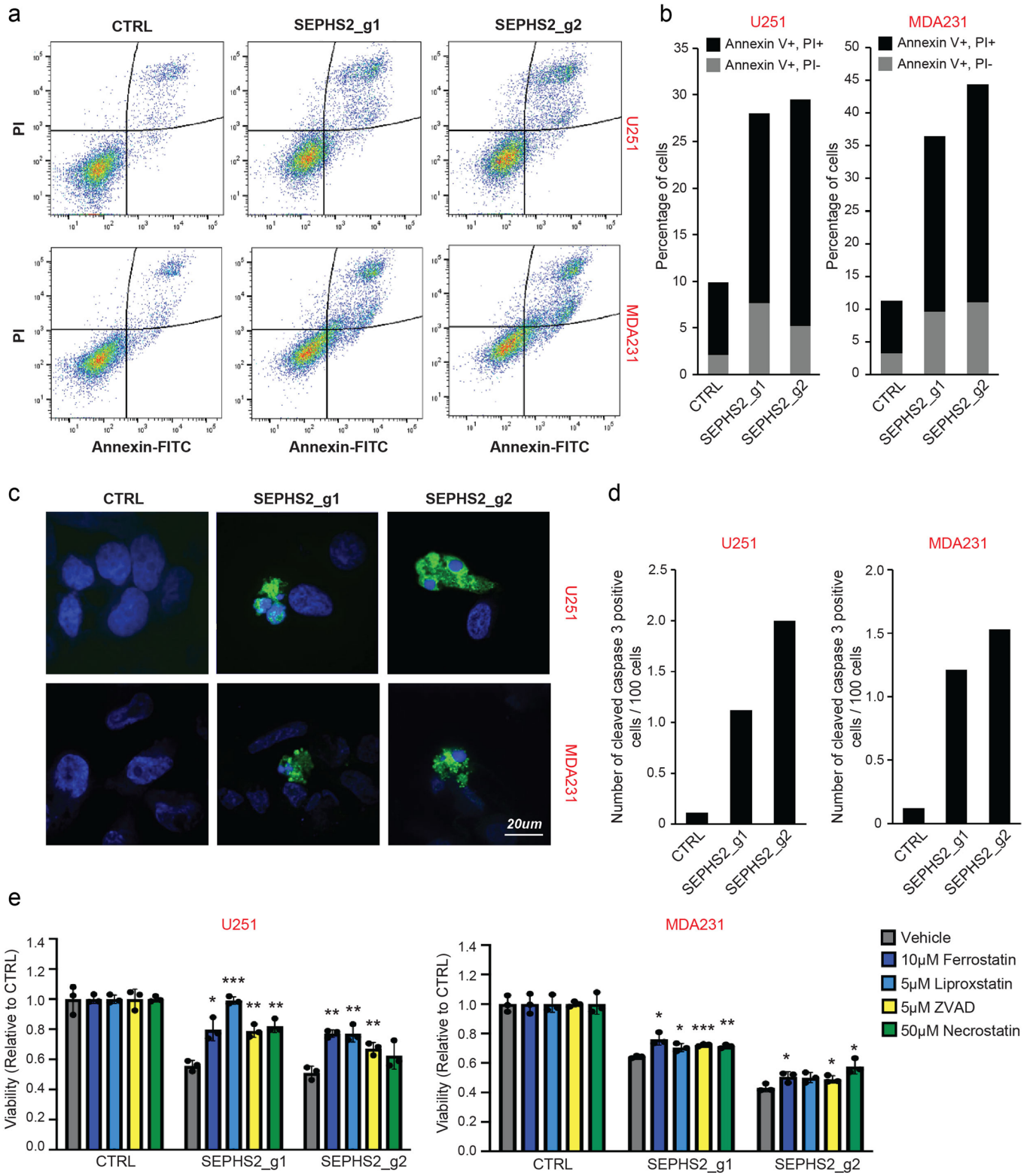
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Toxic metabolite	Chemical structure	Proof of compound toxicity (PMID)	Pathway	Detoxifying enzyme
Cysteamine		2253283, 8616813	Taurine/hypotaurine metabolism	ADO
Inosine monophosphate		7271444	Purine metabolism	ADSS, IMPDH2
Arachidonic acid, Linoleic acid		7898670, 1582681, 11042099	Arachidonic acid metabolism	ALOX15B
2-hydroxyestradiol		15513894	Tyrosine metabolism	COMT
2'-deoxyuridine 5'-triphosphate		2015598	Pyrimidine metabolism	DUT
Glycerophosphocholine		9202319	Glycerol metabolism	GDPD5
Methylglyoxal		25855294	Glycine metabolism	GRHPR
Nicotinamide		4271091	Nicotinate and nicotinamide metabolism	NAMPT
2-Phosphoglycolate		27731369	Glyoxylate and dicarboxylate metabolism	PGP
Retinal		19553623	Retinol metabolism	RDH8
Hydrogen selenide		7357982	Selenocompound metabolism	SEPHS2

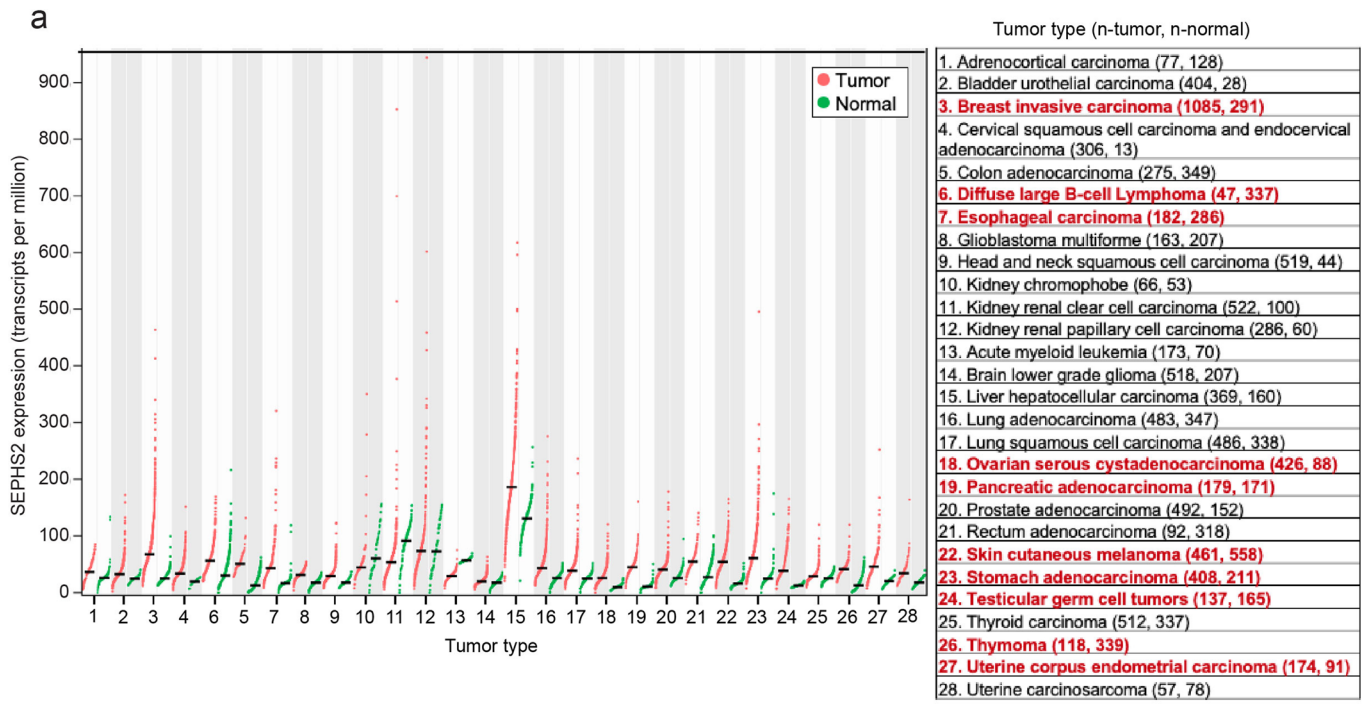
Extended Data Fig. 1 | Panel of toxic metabolites and putative detoxifying enzymes tested. A panel of metabolites that have documented toxic properties, and the downstream enzymes which use them as substrates, designated as 'detoxifiers', are listed. Chemical structures and PMID of references which document the toxicity of each metabolite as a compound are shown.



Extended Data Fig. 2 | Further details of effects of SEPHS2 KO and overexpression in cells and in the MDA231 orthotopic xenograft. a, Viability of cell lines following KO with guides against SEPHS2 for 9 days. Values are relative to the same cell lines expressing nontargeting (CTRL) guide (=1.0), (**b**) Light microscope images of MDAMB231 and MCF10A cells subjected to CTRL or SEPHS2 KO. Scale bar=25μm. **c**, Viability of cell lines following KO with guides against SEPHS2 for 11 days. Values are relative to the same cell lines expressing nontargeting guide (=1.0). **d**, Map of wild type SEPHS2 and CRISPR resistant mutant SEPHS2_U60C. Details of features are provided in Methods. **e**, Viability of vector and SEPHS2_U60C overexpressing cell lines subjected to CTRL and SEPHS2 g1 KO. **f**, Weight measurements for ex vivo CTRL and SEPHS2 KO MDAMB231 orthotopic xenograft tumors. **g**, In vivo tumor volume measurements for CTRL and SEPHS2 KO MDAMB231 orthotopic xenograft tumors. For f and g, if no tumor has formed, tumor weight and volume was designated 0 for CTRL, n = 6 and for SEPHS2 KO n = 2. **h**, Immunoblot of SEPHS2 in CTRL and SEPHS2 KO MDAMB231 orthotopic xenograft tumors. Number is an animal identifier number. For a,c,e, n = 3 biological replicates; error bars are S.D. Number is an animal identifier number. **i**, Viabilities of noncancer (immortalized) and cancer lines following KO with guides against SEPHS1 (blue bars) for 9 days. Values are relative to the same cell lines expressing nontargeting guide (black bars), set at 1.0. **j**, Immunoblots of selenoprotein expression in MCF10A and MDAMB231 cells subjected to (CRISPR/Cas9-induced) KO with guides against SEPHS1, SEPHS2, or control guides for 11 days. For a,c,e and i, n = 3 biological replicates; error bars are S.D. For all panels, the measure of center is mean. *p > 0.05, **p < 0.01 (student's two-tailed t test).



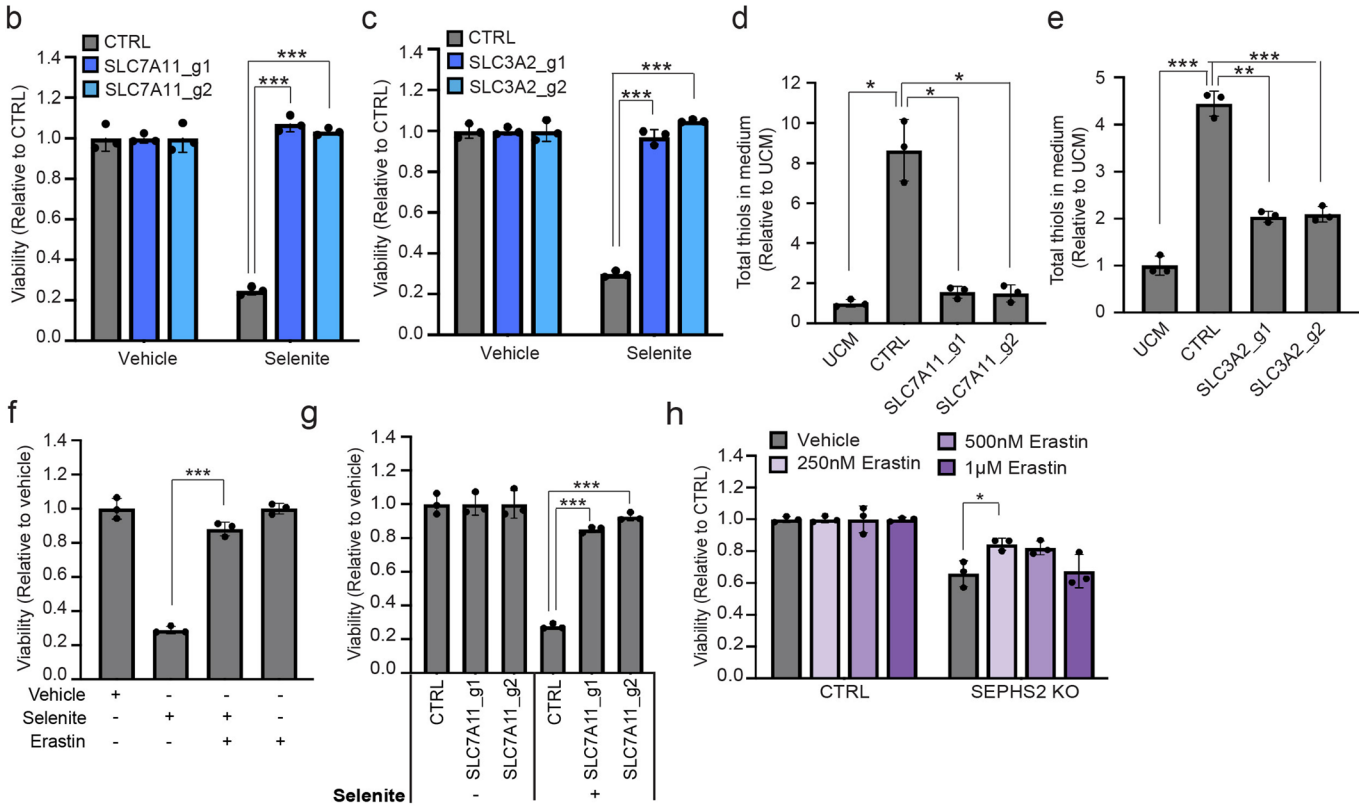
Extended Data Fig. 3 | Induction of cell death by SEPHS2 KO in cancer cells. a, Raw flow cytometry of PI and Annexin V double-stained U251 and MDAMB231 cells following KO with guides against SEPHS2 at 10 days after viral infection. Gating strategy is outlined in Extended Data Fig. 10a–e **b**, Percentages of Annexin V+/PI- and Annexin V+/PI+cells. **c**, Cleaved caspase 3 (CC3) staining in MDAMB231 and U251cells following KO with guides against SEPHS2 at 10 days after viral transduction. Scale bar=20μm. **d**, Quantification of CC3 positive cells over total cells in CTRL and SEPHS2 deficient MDAMB231 and U251. Counts are from random fields. **e**, Viability of control and SEPHS2 deficient U251 and MDAMB231 cells treated with vehicle or the indicated drugs for 4 days. Values are relative to the cell lines expressing nontargeting (CTRL) guide, set at 1.0. For e, n=3 biological replicates; error bars are S.D. For all panels, the measure of center is mean. For all panels, *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$ (student's two-tailed t test).



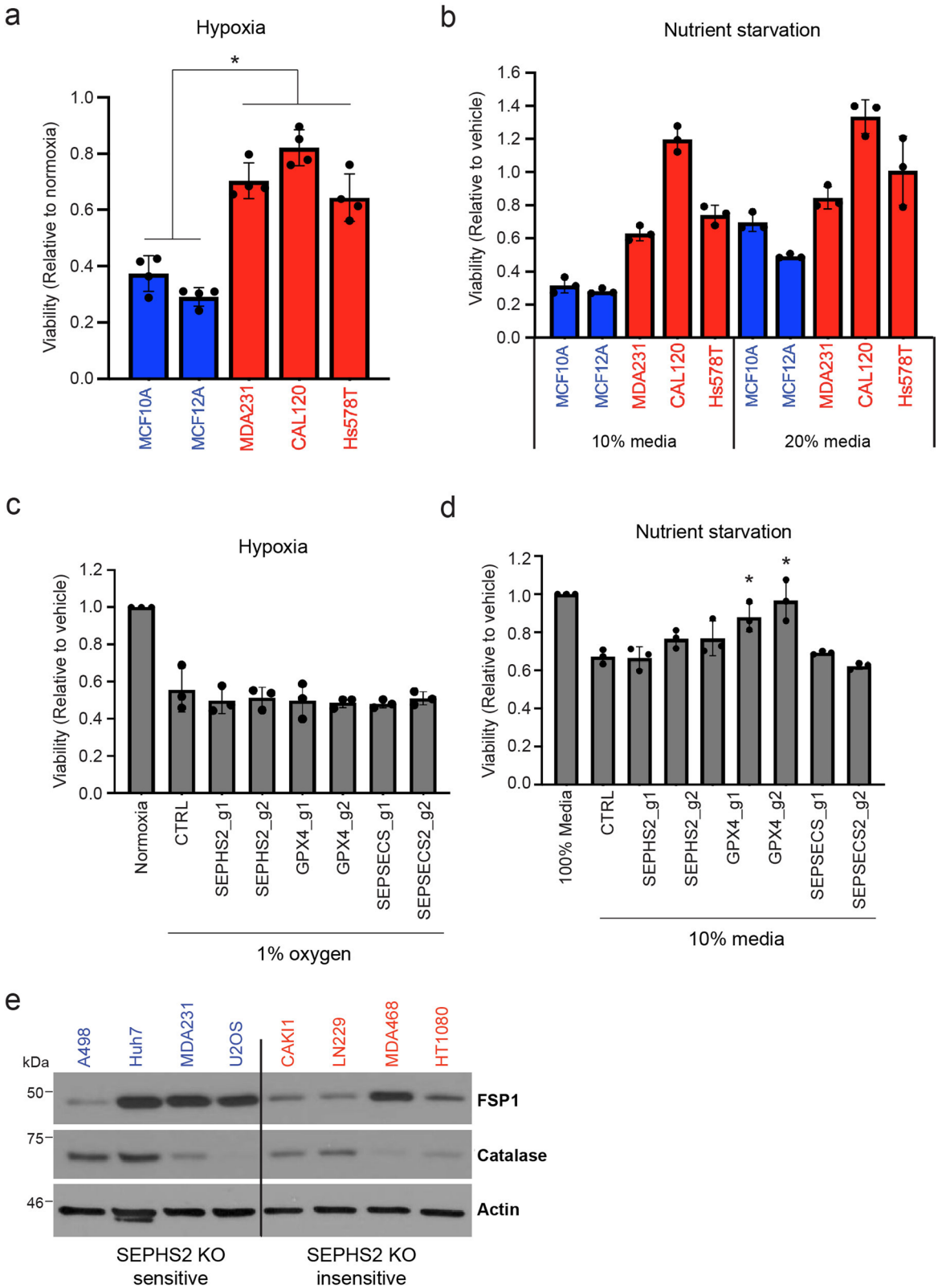
Extended Data Fig. 4 | Expression of SEPHS2 in various types of tumor and normal tissues. a, Expression profiles of SEPHS2 in 28 types of normal and tumor tissues. Normalized mRNA-Seq expression profile data from patient tumor tissues and normal tissues was obtained from GEPIA (Tang, Z. et al. (2017) GEPIA: a web server for cancer and normal gene expression profiling and interactive analyses. Nucleic Acids Res, 10.1093/nar/gkx247). Tumor subtypes having over 2-fold SEPHS2 expression in tumor than normal and q-value less than 0.01 indicated with red. q values are have been determined by ANOVA and adjusted for FDR (false discovery rate). (**b** and **c**) Box plots of SEPHS2 mRNA expression levels in patient tumors and normal breast tissue. SEPHS2 expression was compared in tumor and normal breast samples from TCGA **b**, and METABRIC (**c**) datasets. For panel b, minimum is non-tumor (-0.239), tumor (-0.6386), maximum is non-tumor (0.1874), tumor (1.0702) and median is non-tumor (0.0014), tumor (0.1402). For panel **c**, minimum is non-tumor (1.10097), tumor (1.26305), maximum is non-tumor (3.85158), tumor (5.67272), and median is non-tumor (1.941515), tumor (2.86621).

a

Candidates	Rationale for being a candidate transporter	Reference (PMID)
IMPDH1	Non-transporter metabolic enzyme used as negative control	N/A
DBH	Non-transporter metabolic enzyme used as negative control	N/A
SLC26A11	Homologue of yeast selenate transporter (Sul1p and Sul2p)	18789529, 12626430
SLC4A1	Reported selenite transporter	8238316, 22580993
SLC4A3	SLC4A1 homologue	8238316, 22580993
SLC7A11	Reported indirect modifier of selenium uptake; enriched in selenite-sensitive lines	19549867
SLC39A10	Homology/functional overlap with SLC39A8	29927321, 27166256
SLC39A14	Homology/functional overlap with SLC39A8	29927321, 27166256
SLC39A6	Homology/functional overlap with SLC39A8	29927321, 27166256
SLC39A8	Reported selenite transporter	29927321, 27166256
SLC26A10	Evidence for overlap between sulfur and selenium transport	23148951
SLC26A4	Evidence for overlap between sulfur and selenium transport	23148951
SLC26A9	Evidence for overlap between sulfur and selenium transport	23148951
SLC26A2	Evidence for overlap between sulfur and selenium transport	23148951
SLC26A5	Homologue of yeast selenate transporter (Sul1p and Sul2p)	18789529, 12626430
SLC1A4	Candidate due to high enrichment in selenite-sensitive cell lines	N/A
STARD8	Candidate due to high enrichment in selenite-sensitive cell lines	N/A
SLC13A3	Homologue of putative selenium transporter in rat; enriched in selenite-sensitive lines	16129486
GABRA3	Candidate due to high enrichment in selenite-sensitive cell lines	N/A
KCNE4	Candidate due to high enrichment in selenite-sensitive cell lines	N/A
ABCB4	Homologue of bacterial selenite/selenate transporter	18789529
SLC26A1	Evidence for overlap between sulfur and selenium transport	23148951
SLC26A7	Evidence for overlap between sulfur and selenium transport	23148951

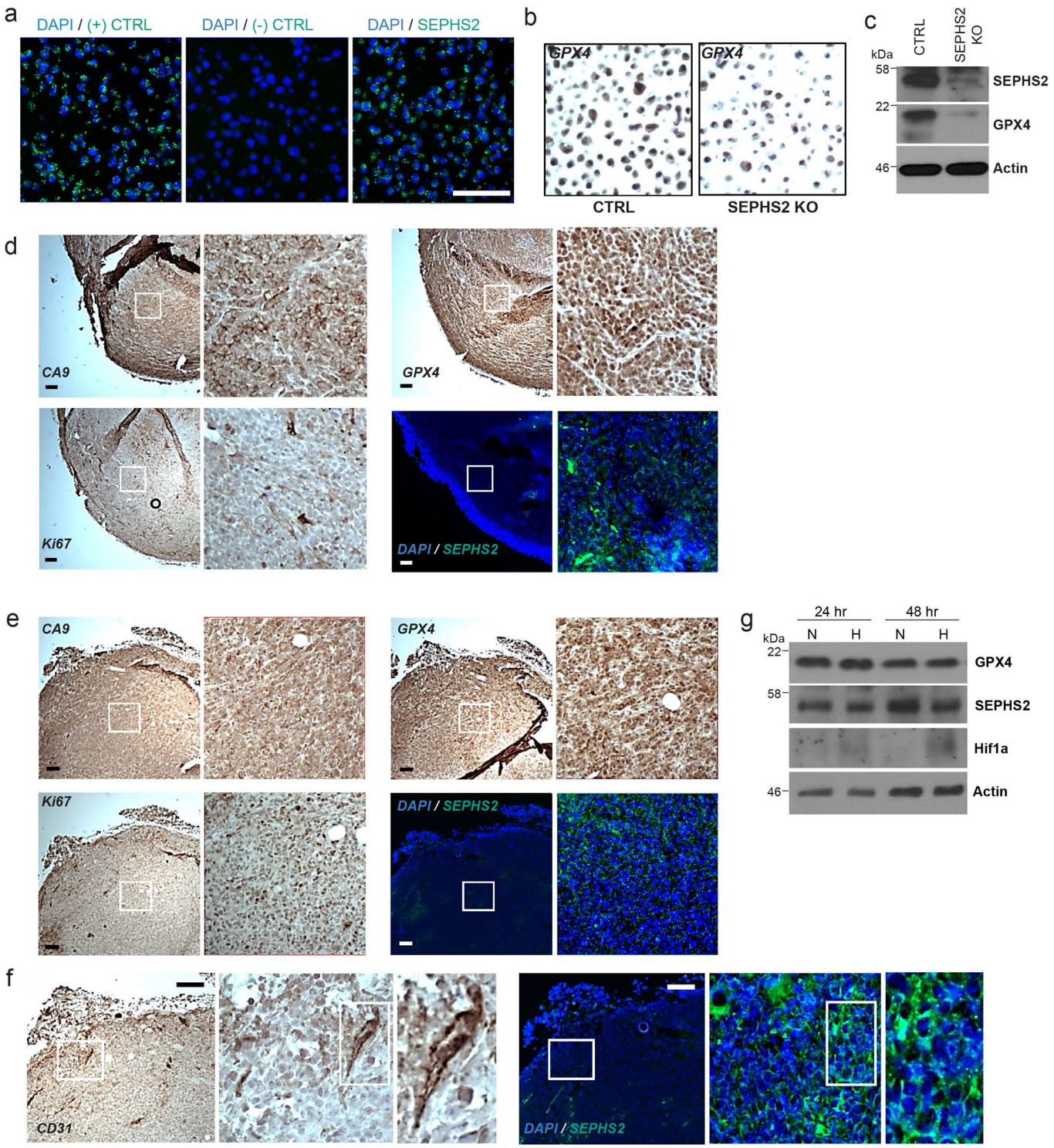


Extended Data Fig. 5 | Candidate transporters tested and xCT functional characterization. a, A list of selenium transporter candidates examined in the focused screen, rationale for candidacy, and references. **b**, Viability of control and SLC7A11 KO U251 cells treated with 12 μM of sodium selenite for 48 hr. Values are relative to that of the same cells treated with vehicle (=1.0). **c**, Viability of control and SLC3A2 KO U251 cells treated with 12 μM of sodium selenite for 48 hr. Values are relative to that of the same cells treated with vehicle (=1.0). **d**, Total thiol quantification of 24 hr conditioned media from control and SLC7A11 KO U251 cells. **e**, Total thiol quantification of 24 hr conditioned media from control and SLC3A2 KO U251 cells. Each value was normalized to that of unconditioned medium. **f**, Viability of U251 cells treated with vehicle, 12 μM sodium selenite, and/or 10 μM Erastin for 72 hr. Values are relative to the cells treated with vehicle (=1.0). **g**, Viability of control and SLC7A11 KO U251 cells treated with vehicle or 12 μM sodium selenite for 72 hr. Values are relative to the control cells treated with vehicle (=1.0). **h**, Viability of control and SEPHS2 KO CAL120 cells treated with vehicle or indicated doses of Erastin for 4 days. Values are relative to each untreated cells (=1.0). For b-h, n = 3 biological replicates; error bars are S.D. For all panels, the measure of center is mean. For all panels, *p < 0.05, **p < 0.01, and ***p < 0.001 (student's two-tailed t test).

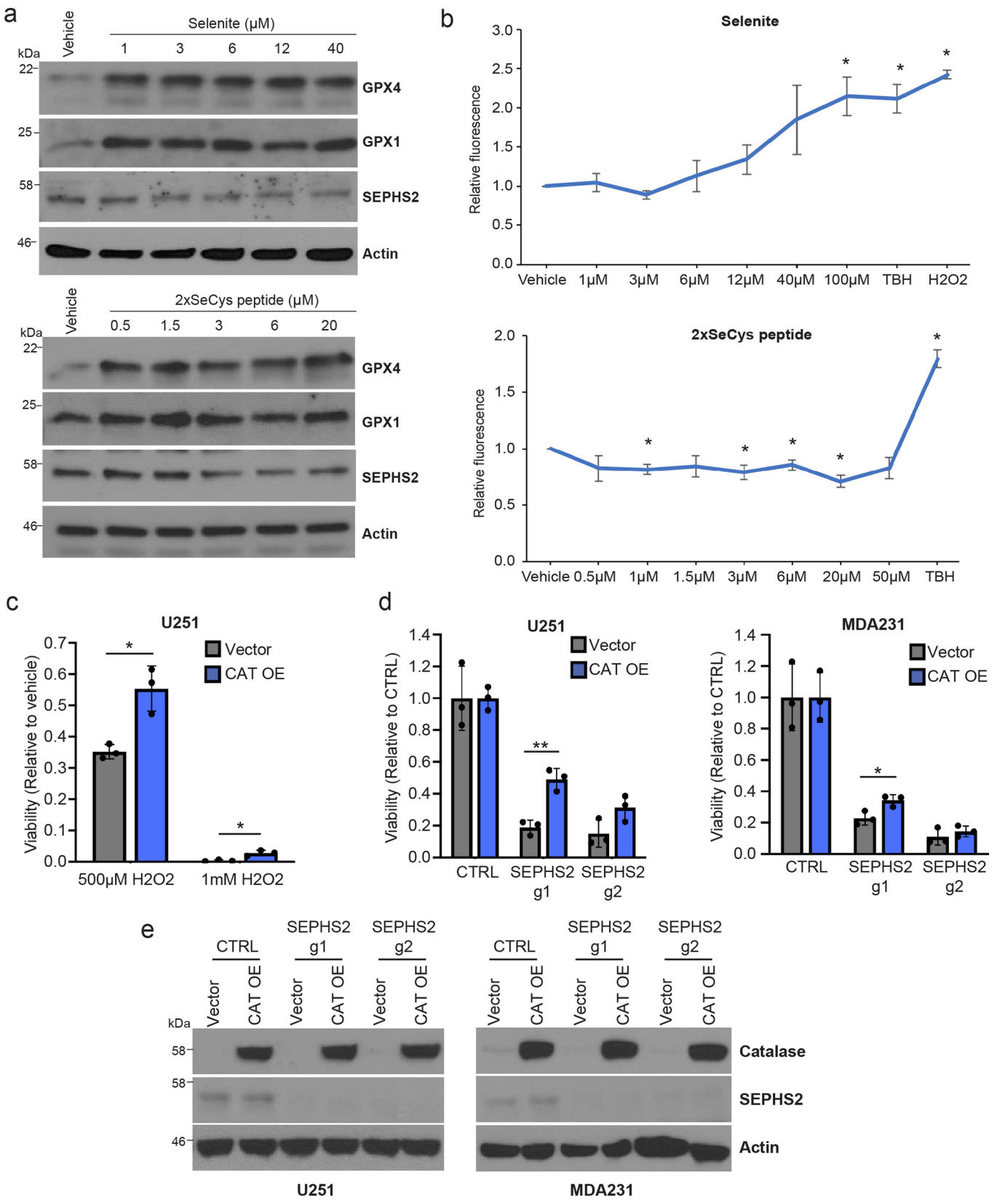


Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Effects of hypoxia or nutrient-starved stresses on normal and selenophilic cancer cells. a, Viability of non-transformed (blue bars) and selenophilic cancer (red bars) cells under hypoxia (1% O_2) for 48 hr. Values are relative to cells cultured under normoxic condition (20% O_2), set at 1.0. **b**, Viability of non-transformed (blue bars) and selenophilic cancer (red bars) cells under nutrient starved conditions (10% or 20% growth media) for 48 hr. Values are relative to cells cultured in 100% growth media, set at 1.0. **c**, Viability of control, SEPHS2, GPX4, and SEPSECS deficient MDAMB231 cells after being cultured in hypoxic condition (1% O_2) for 48 hr. Values are relative to the cells cultured in normoxic condition (20% $O_{2)}$, set at 1.0. **d**, Viability of control, SEPHS2, GPX4, and SEPSECS deficient MDAMB231 cells under nutrient starved conditions (10% growth media) for 48 hr. Values are relative to the cells cultured in 100% growth media, set at 1.0. **e**, Immunoblot of FPS1 and catalase in SEPHS2 KO sensitive and in sensitive cell lines. For **a**,**b**, n = 3 biological replicates; error bars are S.D. For **c**,**d** n = 3 independent experiments; error bars are S.D. For b, while trends can be seen, statistical comparisons were not carried out due to variability between cell lines. For all panels, the measure of center is mean. For all panels, *$p < 0.05$ (student's two-tailed t test).

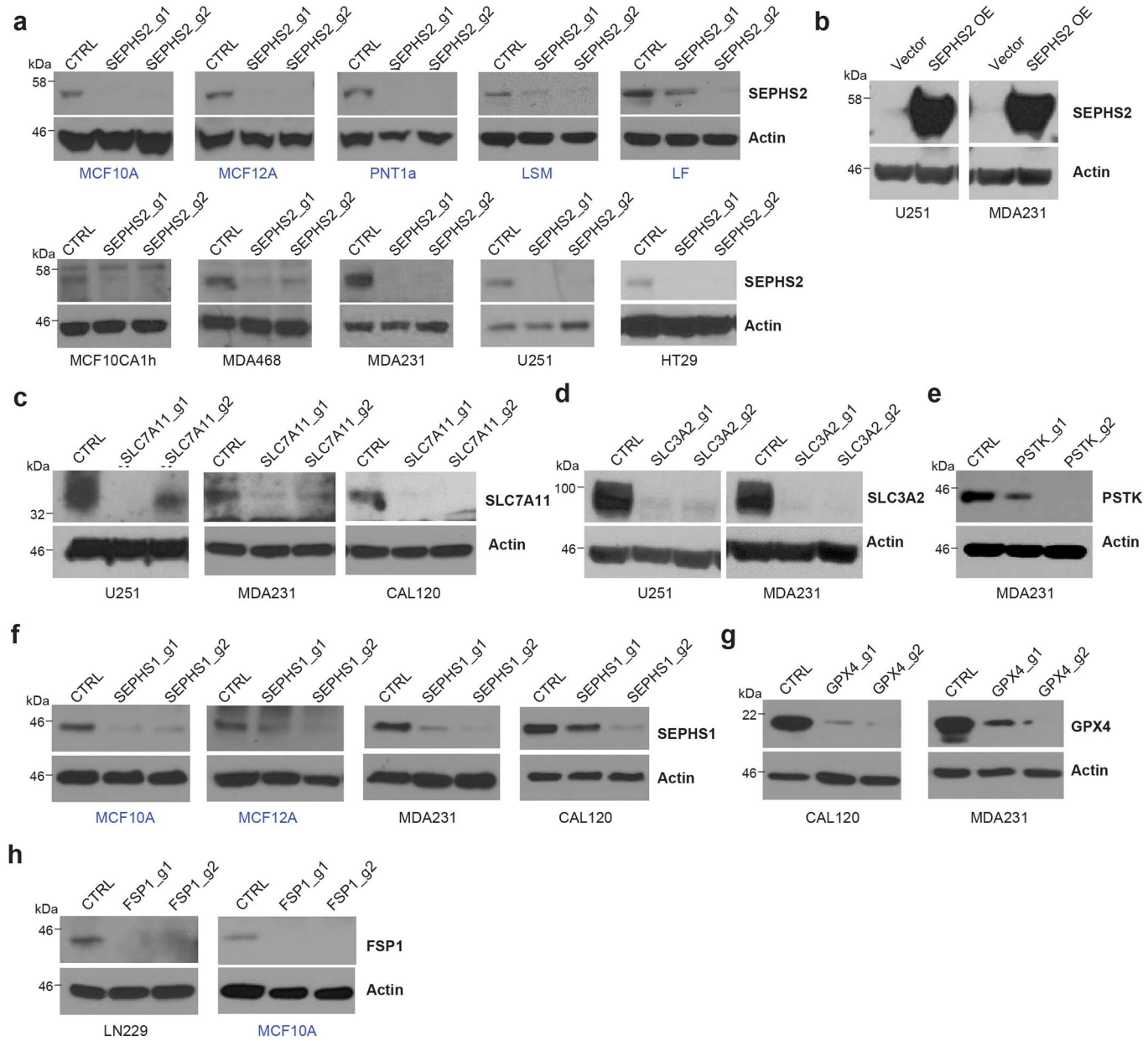


Extended Data Fig. 7 | SEPHS2 and GPX4 expression in tumor microenvironments. a, Validation of SEPHS2 RNA probe by RNAscope in situ hybridization in MDA231 cell pellet block samples. Scale bar=100μm. **b**, Validation of specificity of GPX4 antibody in CTRL and SEPHS2 KO MDAMB231 cell pellet block by immunohistochemistry. **c**, SEPHS2 immunoblots from the cells used in **b**. (**d** and **e**) Micrographs of serial xenograft tumor sections stained with CA9, GPX4, KI67 antibodies and in situ hybridized with SEPHS2 RNA probe. The right hand panels are magnifications of the regions outlined by the white square. **d**, shows a hypoxic region as indicated by membrane staining of CA9, and with a low Ki67 index. **e**, shows a nonhypoxic region (as indicated by only background nonspecific stain in CA9) with a high Ki67 index. Scale bar is 100μm. **f**, Micrographs of serial human tumor sections stained with CD31 antibody and in situ hybridized with SEPHS2 RNA probe, showing vasculature. Scale bar is 100μm. **g**, Immunoblots of GPX4, SEPHS2 and Hif1a in MDAMB231 cells subjected to hypoxia (1% O_2) for 48hrs.

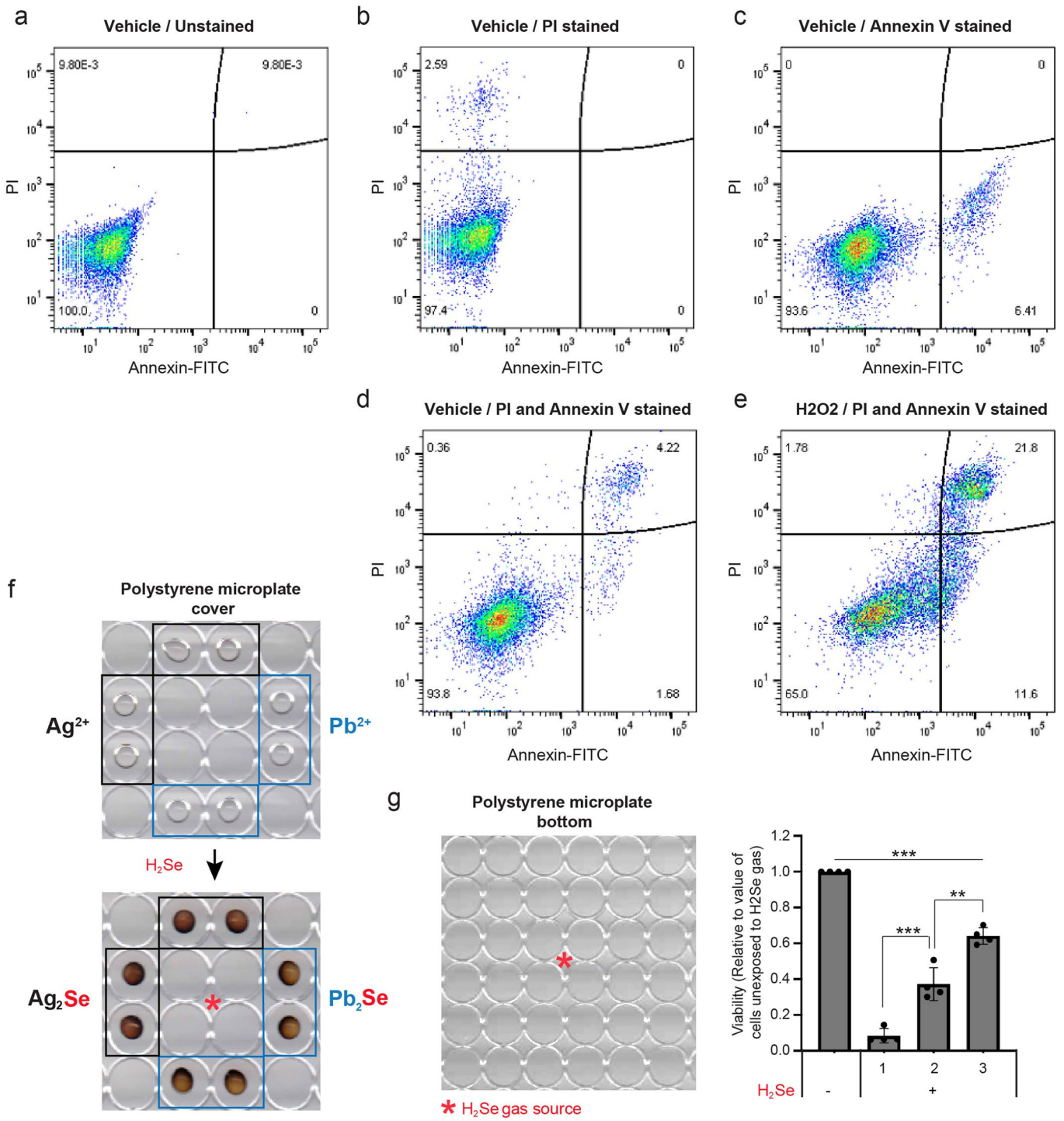


Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Reactive oxygen species production by excess selenium, and implications in SEPHS2 KO toxicity. a, Selenoprotein immunoblots of MDAMB231 cells at 24 hr recovery after 2 hr sodium selenite treatment or treated with 2-selenocysteine-containing peptide for 24 hr. **b**, ROS quantification of cells treated with indicated concentrations of sodium selenite or 2xselenocysteine containing peptide, 250 μM TBH, or 500μM H_2O_2 for 45 mins. **c**, Viability of U251 cells overexpressing vector or CAT treated with vehicle or H_2O_2 for 48 hr. **d**, Viability of U251, and MDAMB231 cells overexpressing blank vector or CAT and subjected to KO with guides against SEPHS2 for 9 days. **e**, Immunoblots of CAT and SEPHS2 in U251 and MDAMB231 cells overexpressing blank vector or CAT and subjected to KO with guides against SEPHS2 for 9 days. For b, n = 3 independent experiments; error bars are S.D. For c,d n = 3 biological replicates; error bars are S.D. For all panels, the measure of center is mean. For all panels, *$p < 0.05$ and **$p < 0.01$ (student's two-tailed t test).



Extended Data Fig. 9 | Confirmation of gene knockout or overexpression after lentiviral transduction. a, Immunoblots of SEPHS2 in 5 non-transformed (blue) and 5 transformed (black) cell lines transduced with Cas9/sgRNAs against SEPHS2, and control (non-targeting sgRNA). **b**, Immunoblots of SEPHS2 in cells transduced with CRISPR resistant SEPHS2_U60C and control (empty vector). **c**, Immunoblots of SLC7A11 in cells transduced with Cas9/sgRNAs targeting SLC7A11 or non-targeting sgRNA. **d**, Immunoblots of SLC3A2 in cells transduced as indicated **e**, Immunoblots of PSTK in cells transduced as indicated. **f**, Immunoblots of SEPHS1 in 2 non-transformed and 2 cancer cells transduced as indicated. **g**, Immunoblots of GPX4 in cells transduced as indicated. **h**, Immunoblots of FSP1 in cancer cells transduced as indicated. All samples were lysed between 7~8 days after viral transduction.



Extended Data Fig. 10 | Supporting data for flow cytometry and gas toxicity system. a–e, Representative flow cytometry data to address gating and quadrant strategy for FACS measurement of dying/dead cells. MDAMB231 cells were stained with Annexin V-FITC and propidium iodide (PI). The X-axis and y-axis of Flow cytometry density plots were designated as Annexin V-FITC and PI, respectively. The left lower quadrant represents Annexin V negative and PI negative live cells. The right lower quadrant represents Annexin V positive and PI negative early dying cells. The right upper quadrant represents Annexin V positive and PI positive late dying/dead cells. Staining is as follows: (**a**) unlabeled, (**b**) PI, (**c**) Annexin V, (**d**) Annexin V and PI, (**e**) Annexin V and PI stained cells treated with 50 μM H_2O_2 for 24 hr. **f**, Scanned image of a polystyrene microplate cover with PVP spots embedded with silver nitrate or lead acetate. The plate cover was scanned before (upper panel) and after (lower panel) selenide gas exposure for one min. **g**, Viability of MDAMB231 cells in the numbered spots outlined on the 96 well plate, exposed to selenide gas. For g, n = 4 independent experiments; values are normalized to untreated cells (=1.0). For all panels, the measure of center is mean. For all panels $^{**}p < 0.01$, and $^{***}p < 0.001$ (student's two-tailed t test).

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

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Data collection	Beckman Coulter DTX880 Multimode Detection Software version 1.0 was used to measure CellTiterGlo-based cell viability and ROS detection in 96 well plates; Thermo Scientific Nanodrop 2000 V1.0 was used in nucleic acid measurement; Gen5 version 3.03: BioRad Image Lab Version 6.0.1 was used for scanning western immunoblots; BD LSR II flow cytometer was used in Annexin V and PI double stain; Lionheart LX (Serial 1710281) Automated microscope was used for taking images of immunocytochemistry and RNA scope.
Data analysis	Microsoft Excel for Microsoft 365 MSO (16.0.12730.20188), Graphpad PRISM version 8, ImageJ version 1.520, FLOWJO version 10, ChemDraw Professional version 18.0, and survival package (v2.42-6) in R 3.5.3 were used for data analysis, statistical data analysis, or for making of chemical structures.

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All raw data that support the findings of this study are available from the corresponding author upon request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Most experiments, in particular experiments involving CellTiterGlo-based viability were carried out in triplicate which we have found in our previous studies to be sufficient for determining effects of pLentiCRISPR-based genetic perturbations and drug treatments on cell viability in cancer cell lines. Mouse orthotopic xenografts were set up with 7 mice per cohort as we have previously obtained statistically significant results with this number of mice using this cell line model in previous experiments that projected a >50% difference in cell viability. Numbers of cancer cell lines and normal cell lines for comparing SEPHS2 effect in cancer vs normal lines were determined by availability of cell lines that were infectable and able to be selected using antibiotics. Human breast tumor and normal breast samples were based on sample availability. Experiments involving volatile selenite gas toxicity were in sets of 4 which were based on convenient equidistant plating in a 96 well format.
Data exclusions	No data was excluded.
Replication	All experiments were repeated at least three times with the following exceptions: Focused candidate detoxifying enzyme screen carried out in 11 cell lines in Fig 1b, effects of SEPHS2 KO on tumor growth (Fig 1f,g and Extended Data Fig 2f,g), focused screen for selenium uptake mediator (Fig 3a), SEPHS2 protein measurements in tumors vs normal tissues (Fig 1h,i), imaging for cleaved caspase 3 in SEPHS2 KO cells (Extended Data Fig 3c,d), selenium ICP/MS quantification of captured spots (Fig 4g), validation of SEPHS2 expression knockout (Extended Data Fig 9), and validations of drug efficacy (Fig 3j and Extended Data Fig 8c) which were performed once. The following experiments were repeated twice: SEPHS2 KO in multiple cell lines (Fig 1c and d), SEPHS1 KO in multiple cell lines (Extended Data Fig 2i), SEPHS2/GPX4/SEPSECS KO cell treatment with hypoxia and nutrient starvation (Extended Data Fig 6c,d) and erastin treatment of SEPHS2 KO (Extended Data Fig 5h). In all experiments, all attempts at replication were successful.
Randomization	For xenograft tumor experiments, Nu-/nu- athymic mice were randomly assigned to be injected with SEPHS2 KO MDAMB231 cells or CTRL KO MDAMB231 cells.
Blinding	For xenograft tumor experiments, the researchers carrying out injection and tumor measurements were blinded to which group were which (SEPHS2 KO vs CTRL KO tumors).

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Antibodies for WB
Rabbit polyclonal anti-SEPHS2 (dilution-1:1000), Sigma-Aldrich, Cat#SAB2700271, Lot#GT39876
Rabbit polyclonal anti-PSTK (dilution-1:1000), Invitrogen, Cat#PA5-58101, Lot#TL2685284
Rabbit monoclonal anti-GPX4 (dilution 1:1000), Abcam, Cat#ab125066, Lot#GR3229900-6
Rabbit polyclonal anti-GPX1 (dilution 1:500), Abcam, Cat#ab22604, Lot#GR3185180-1
Rabbit monoclonal anti-xCT/SLC7A11 (dilution-1:1000), Cell signaling Technology, Cat#12691, Lot#1
Rabbit polyclonal anti-SLC3A2 (dilution-1:2000), Sigma-Aldrich, Cat#HPA017980, Lot#A115302
Rabbit polyclonal anti-SELS (VIMP) (dilution 1:500), Sigma-Aldrich, Cat#HPA048362, Lot#A101692
Rabbit polyclonal anti-SELH (C11ORF31) (dilution-1:1000), Sigma-Aldrich, Cat#HPA010025, Lot#R58332
Rabbit monoclonal anti-Catalase (D4P7B) (dilution-1:1000), Cell signaling Technology, Cat#12980T, Lot#2
Mouse monoclonal anti-SEPHS1 (F-6) (dilution-1:200), Santa Cruz, Cat#sc-365945, Lot#D1718

Mouse monoclonal anti-AMID (B-6) (dilution-1:200), Santa Cruz, Cat#sc-377120, Lot#A2419
Mouse monoclonal anti-β-Actin (dilution-1:5000), Sigma-Aldrich, Cat#A2228, Lot#026M4780V
Rabbit monoclonal anti-β-Actin (13E5) (dilution-1:2000), Cell signaling Technology, Cat#4970, Lot#18
Antibodies for ICC
Rabbit polyclonal anti-Cleaved Caspase-3 (Asp175) (dilution-1:200), Cell signaling Technology, Cat#9661, Lot#45
Goat anti-Rabbit IgG H&L (Alexa Fluor® 488) (dilution-1:200), Abcam, Cat#ab150077, Lot#GR3280242-1
Rabbit monoclonal anti-GPX4 (dilution-1:200), Abcam, Cat#ab125066, Lot# GR3229900-6
Rabbit polyclonal Anti-CA9 (dilution-1:200), Sigma-Aldrich, Cat#HPA055207, Lot#000000565
Mouse monoclonal PECAM-1 Antibody (10G9) (dilution-1:200), Santa Cruz, Cat#sc-13537, Lot#A0716
Mouse monoclonal anti-MKI67 (dilution-1:200), Sigma-Aldrich, Cat#AMAB90870, Lot#02780

Validation

Most of the antibodies listed above were validated by CRISPR KO or target gene OE to compare bands from lysates with and without the protein. Such KO validation data (i.e. loss of predicted bands in the KO cells) is shown in the manuscript for SEPHS2, PSTK, GPX4, xCT/SLCL7A11, SLC3A2, FSP1 and SEPHS1, in human cells for immunoblots. KO validation for GPX1, SELS and SELH is not shown in the manuscript data. Overexpression validation for anti-Catalase antibody is in the manuscript. The anti-Actin antibody used as loading control is widely used and validated in the research community. The anti-CA9, anti-PECAM-1 and anti-MKI67 antibodies are widely used and validated in the research community as markers for hypoxic, endothelial, and proliferating cells, respectively; these showed the expected staining pattern in human paraffin section IHCs. The anti-cleaved caspase 3 antibody is widely used as a marker for imaging of apoptotic cells, and showed the expected staining pattern in immunocytochemistry of human cells.

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)

LF and LSM primary lines were obtained from ATCC. THLE2, CCD18CO, MCF10A, and HUH7 were a kind gift from the Kwan Yong Choi lab (POSTECH, KOREA). PNT1A were a kind gift from the Arthur Mercurio lab (UMMS, USA). MCF12A and MCF10CA1h were a kind gift from the Michael Green lab (UMMS, USA). U2OS, U251, LN229, CAL120, MDAMB415, MDAMB468, MDAMB231, HS578T, SW620, HT1080, A498, CAKI1, A2058, A549, WM115 and DU145 were a kind gift from the David Sabatini lab. U118/138 were a kind gift from Matija Snuderl (NYU, USA). SUM159 were a kind gift from the Leslie Shaw lab (UMMS, USA). MDAMB157, HCC1395, and MDAMB436 were a kind gift from the Michael Lee lab (UMMS, USA).

Authentication

Please refer to Table S2 in the manuscript for a detailed list. All the cell lines used in the study were authenticated via STR profiling of 6 loci with the exception of LF and LSM (purchased from ATCC in December 2018) and Sum159, MDAMB436, and HCC1395 (not profiled). Additionally, in the PNT1A cell line 4/4 loci matched but reference data was not available for other loci that we tested. STR profiling was performed at the Duke University DNA Facility for all cell lines except U2OS, CAL120, HS578T, A498 and A2058, WM115, A549 which were previously STR validated in the lab from which we obtained them (Sabatini lab).

Mycoplasma contamination

MDAMB231, U251, LN229, HT29, MCF12A, and CCD18CO cell lines were confirmed to be mycoplasma negative using e-Myco Mycoplasma PCR Detection Kit, catalog number 25235 from iNtRON Bio. Other cell lines have not been tested for mycoplasma.

Commonly misidentified lines
(See ICLAC register)

One of the lines U118, a glioma line, is thought to be the same line as another glioma line (U138) used in the research community. We have STR validated this as being U118/138 and this does not impact our findings in any way. We note this feature of U118 in Supplemental Table 2.

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals

8-week-old female athymic nu-/nu- mice (mus musculus)

Wild animals

The study did not involve wild animals.

Field-collected samples

The study did not involve samples collected from the field.

Ethics oversight

Institutional Animal Care and Use Committee (IACUC) at University of Massachusetts Medical School

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about studies involving human research participants

Population characteristics

Frozen breast cancer tumor and matched normal adjacent breast tissues were obtained from female patients, ranging in age from 46 to 74 years old. Further deidentified details can be provided upon request.

Recruitment

Samples were harvested during tumor resection at the clinical partner of the University of Massachusetts Medical School, UMass Memorial Health Care in Worcester, Massachusetts.

Ethics oversight	The samples were collected in accordance with ethical approval by the UMass Medical School Institutional Review Board (Docket # H-11731).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Sample preparation follows FITC Annexin V Apoptosis Detection Kit (BD Pharmingen, Cat#556547). Briefly, cells were washed with cold PBS and resuspended in 1X binding buffer with Annexin V-FITC and propidium iodide (PI) at room temperature for 15 min, and then analyzed immediately.
Instrument	BD LSR II flow cytometer was used.
Software	BD FACSDiva program was used for data collection and FLOWJO 10 was used for data analysis.
Cell population abundance	Around 90% cells were gated as a single cells without the cell debris from the full population. Around 10000 single cells were used to sort and analysis.
Gating strategy	FSC-A and SSC-A gating strategy was used to eliminate the cell debris. FSC-H and FSC-A subgating strategy was used to identify single cell population. For Annexin V-FITC positive and Annexin V-FITC/PI positive cells were gated for early and late dying/dead cells.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.